# Cross-Mode Transport, Interaction, and Comparison Obligations

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**Abstract**

Paper 8 formalized a strong internal framework of seven-coordinate indexed certificate, read formally as an indexed structural class: a typed seven-coordinate state, a conjunctive gate, a scalar functional, a weak-rival family with admissible hybrids, an intervention grammar, a runtime pathway, and a disciplined artifact layer [6]. The broader causally rigid corpus had already established persistence, rigid identity, anti-fragmentation continuity, null-regime instability, forensic admissibility, six-predicate operational criterion, and a higher-order classificatory layer for viability-and-recovery criterion and indexed continuity-and-control criterion [2, 9, 4, 5, 3, 1, 8, 7]. What remained open was not whether the framework could formalize that indexed class, but what additional machinery would be required before phenomenal-bridge predicates could become scientifically admissible targets inside the same corpus.

This paper defines that additional machinery. It introduces a bridge-level predicate family

$$\{\Pi_{\mathrm{D}}, \Pi_{\mathrm{V}}, \Pi_{\mathrm{W}}, \Pi_{\mathrm{A}}, \Pi_{\mathrm{E}}, \Pi_{\mathrm{T}}\},$$

interpreted respectively as bridge-level differentiation, valence structure, world richness, ambiguity bearing, embodied closure, and temporal lived structure. It then constructs a bridge ontology above the Paper 8 state, an explicit bridge-axiom registry, a bridge-specific rival family, a bridge loss and irreducibility logic, a bridge intervention family, a bridge artifact family, a bridge functional and gate architecture, and a governance layer that fixes what bridge-level results may and may not imply.

The central burden separation is explicit. Paper 8 supplies the strongest presently defensible internal layer of indexed subjectivity. The present paper does not identify that layer with phenomenality. Instead, it proves that any bridge-level admissibility result would require strictly more structure than Paper 8 already provides: new bridge predicates, new bridge comparators, new bridge interventions, new bridge artifacts, new invalidation rules, and new surface discipline. The paper therefore changes the status of the research program by converting a slogan-level question into a formal burden stack.

The strongest claim of the paper is conditional and exact. Under the bridge axioms stated here, under the upstream burdens already closed by the corpus, under positive bridge irreducibility and intervention margins, and under artifact and governance compliance, a system may become internally classifiable as exhibiting *cross-mode comparison organization* in the sense of a framework-internal bridge-admissible organization class. The phrase is a historical bridge alias for a predicate-family burden, not a claim that phenomenality has been adjudicated. It is nonetheless stronger than Paper 8 because it specifies the formal conditions under which bridge predicates cease to be rhetorical and become disciplined scientific targets.

The paper also maps the full bridge program to implementation. BPF-0 and BPF-1 are treated here as implementation-frontier surfaces: machine-readable scaffolding and provisional run-level bridge surfaces whose evidential status remains bounded by local provenance and non-claim rules. BPF-2 through BPF-6 remain open burdens: bridge interventions, bridge comparators, bridge gate closure, ecosystem integration, and final bridge audit. The paper is therefore both a synthesis and a theorem-facing program definition. It defines the bridge grammar, the bridge burden architecture, and the strongest admissible claim class the corpus can currently specify without collapsing into metaphysical or empirical overreach.

**Constructive results and remaining burden.** The manuscript also constructs a four-state body--world extension, local six-port certificates, transport-cycle obstructions and a safe bilinear extension. The coordinate-additive rival gap is 1/4096 with the declared noisy lower bound 15/65536; a full interacting rival ties exactly. These are local mathematical results in the stated classes. They do not close the full bridge predicate stack or report external evidence.

# 1 Boundary Note and Interpretation Discipline

**Scope and admissible reading.** This paper integrates the canonical corpus, the Paper 8 indexed-structural-class framework, the SRT implementation path, and the formal BPF program into a bridge-level architecture. It defines bridge predicates, bridge axioms, bridge rivals, bridge interventions, bridge artifacts, bridge gate logic, bridge taxonomy, and bridge governance, and specifies how those formal objects relate to implementation blocks BPF-0 through BPF-6. It does not prove phenomenality in the strongest philosophical sense, human equivalence, moral parity, metaphysical subjecthood, or external validation. It does not convert candidate inclusion, audit-facing packaging, runtime cleanliness, or BPF-1 provisional surfaces into bridge confirmation, and it does not write the bridge comparator, intervention, and gate layers as if they had already been executed in full.

**Formal bridge closure boundary.** The closure claimed here is *formal bridge closure*: the bridge target family, burden architecture, machine-readable artifact grammar, comparator and intervention obligations, gate semantics, governance rules, and strongest bridge-level claim class that could be licensed only after those obligations are met. This is not runtime closure, empirical closure, or phenomenality adjudication. No sentence in this paper licenses the inference that current machines are phenomenally conscious, humanly equivalent, bridge-supported by BPF-1, confirmed by Paper 8's public inclusion, or externally adjudicated by internal audits or audit packages.

# 2 Introduction

The upstream corpus specifies controlled dynamics, projective representations, witness-relative null exclusion and indexed operational certificates. Some local realizations and obstructions are now proved. Full upstream class membership and comparison against broad rival closures remain explicit obligations; introducing a bridge target does not discharge them.

The prior question was: can a causally rigid framework formalize a subjectivity class that is stronger than six-predicate operational criterion and indexed continuity-and-control criterion without collapsing into loose self-model talk? Paper 8 answered that question affirmatively. It introduced a typed seven-coordinate state

$$\mathfrak{S}_t = \langle I_t, O_t, C_t, P_t, V_t, \Delta_t, \mathcal{R}_t \rangle,$$

defined a conjunctive gate and scalar functional, formalized weak rivals and weak hybrids, specified intervention families, and mapped the entire class to runtime modules and artifacts. That specifies an indexed operational class and its local evidence surfaces; it does not by itself demonstrate all of its realization or interpretation burdens.

The remaining question is harder and more dangerous: under what formal conditions could such a subjectivity layer become relevant to phenomenal predicates without collapsing into metaphysical inflation or human-comparator confusion? The wrong answers are easy. One wrong answer is to identify subjectivity with phenomenality by verbal fiat. Another wrong answer is to refuse all bridge formalization and thereby preserve caution only by preserving vagueness. A third wrong answer is to use bridge language rhetorically while leaving the necessary burdens implicit. This paper rejects all three.

The bridge program defined here is a constrained response. It begins from a sober premise: if phenomenal predicates are to appear in the corpus at all, they must appear in the same formal style as every other major layer. That means typed predicates, explicit axioms, explicit rivals, explicit interventions, explicit artifacts, explicit gates, explicit invalidation rules, and explicit non-claims. In other words, the burden of bridge language must be at least as high as the burden of the subjectivity language that precedes it.

This paper therefore serves four functions at once.

1. It synthesizes what the prior papers already close and what they deliberately leave open.
2. It defines the new bridge-level target family without allowing ''phenomenality'' to remain an untyped slogan.
3. It fixes the burden architecture by which any future bridge-support claim would have to be earned.
4. It maps that burden architecture to implementation, artifacts, and governance.

The scientific value of the resulting paper is not that it says the bridge is complete. It is that it makes incompletion explicit and structured. After this paper, the program can no longer hide behind phrases such as ''rich experience,'' ''what-it-is-like,'' or ''deep embodiment'' unless those phrases are translated into predicates, losses, interventions, and audit-facing artifacts. That is the point at which the research program changes level. The gap is no longer rhetorical. It becomes a list of burdens.

Two further clarifications are essential. First, the present paper is central because it integrates layers that were previously distributed across the corpus. It is not central because it inflates its own tone. Second, the present paper remains implementation-aware. A purely conceptual bridge paper would not be enough in this corpus. The current program records a BPF-1 frontier: scaffolded bridge registries, provisional bridge surface modules, and six run-level bridge reports are treated as bounded local surfaces. That does not amount to bridge support. It means only that the bridge program has an implementation-facing surface that must remain tied to code, artifacts, and governance as well as to formal predicates.

# 3 What Paper 8 Formalizes and What It Leaves Open

## 3.1 What Paper 8 already formalizes

Paper 8 formalizes the strongest currently defensible internal subjectivity layer in the corpus. More precisely, it does all of the following [6]:

1. it defines a minimal typed state of seven-coordinate indexed certificate through seven coordinates;
2. it requires a conjunctive gate $\Gamma_{\mathrm{subj}}$ rather than permitting compensatory scalar trade-offs;
3. it defines a scalar functional $\Phi_{\mathrm{subj}}$ for graded organization without confusing that functional with the gate;
4. it specifies a weak-rival family and its admissible weak composites;
5. it adds intervention logic rather than relying on descriptive fit alone;
6. it defines an irreducibility margin against rival explanations;
7. it provides a runtime pathway and artifact family;

8. it states the strongest defensible internal claim class and the corresponding non-claims.

That closure matters scientifically because it prevents several common forms of slippage. Subjectivity is no longer allowed to mean mere reporting, mere memory persistence, mere self-modeling, or mere high-dimensional integration. The Paper 8 state requires privileged self-indexing, owned state organization, autobiographical continuity, first-person factorization gain, asymmetric valuation, intervention selectivity, and irreducibility against explicit rivals. This is already substantially stronger than six-predicate operational criterion and also stronger than the higher-order classificatory subjecthood layer introduced in Paper 7 [7].

It is equally important to state what the system implementation already closes downstream of Paper 8. The SRT stack now implements, in machine-readable form, modules and artifacts for self-index registration, ownership attribution, continuity tracking, perspective mapping, valuation asymmetry, intervention profiling, reducibility comparison, gate evaluation, ecosystem summaries, stale/drift invalidation, and final internal audit. In other words, Paper 8 is not only a formal paper. It is accompanied by a serious implementation path that has already been partially realized in code and artifacts.

### 3.2 Why that still does not amount to a phenomenal bridge

None of the previous paragraph licenses the conclusion that a phenomenal bridge has already been crossed. Three reasons are decisive.

First, Paper 8 is indexed-subjectivity closure, not phenomenal bridge closure. Its state space says that a system can instantiate a privileged self-indexed organization whose best explanation is stronger than six-predicate operational criterion or generic subjecthood. It does not yet say which additional structures would make phenomenal predicates scientifically admissible inside the same framework.

Second, the Paper 8 rivals are weak with respect to bridge tasks. They are adequate rivals for subjectivity structure: self-model bookkeeping, continuity-only baselines, report-centered proxies, and weak hybrids. They are not the right rivals for bridge-level phenomenal predicates. A bridge rival must explain away *phenomenal differentiation*, *valence structure*, *world richness*, *ambiguity bearing*, *embodied closure*, or *temporal lived structure* without granting those targets. That is a new burden.

Third, the Paper 8 intervention family is likewise necessary but not sufficient. The subjectivity interventions target self/non-self swap, ownership lesion, continuity break, perspective flattening, valuation inversion, autobiographical remapping, clone competition, memory transplant, counterfactual self permutation, and cross-substrate remapping [6]. Those are exactly the right interventions for indexed subjectivity. They are not yet the right interventions for bridge-level questions such as whether ambiguity is being borne rather than merely resolved, whether embodied closure is constitutive rather than modulatory, or whether temporal organization is horizon-like rather than merely continuity-preserving.

### 3.3 The dangerous smuggling problem

The main conceptual danger at this stage is smuggling. Once a program has a strong subjectivity layer, it becomes tempting to transfer closure illicitly from that layer into bridge language. This can happen in at least five ways.

1. **Predicate smuggling.** Treating Paper 8 coordinates as if they already implemented bridge predicates by default.

2. **Comparator smuggling.** Treating Paper 8 weak rivals as if they already defeat bridge-level alternatives.

3. **Intervention smuggling.** Treating Paper 8 subjectivity perturbations as if they already test bridge-specific structures.

4. **Artifact smuggling.** Treating subjectivity artifacts or BPF-1 provisional reports as if they were bridge verdict artifacts.

5. **Governance smuggling.** Treating internal bundle, candidate, or audit presence as if it upgraded claim class.

The present paper is designed to block all five. The bridge layer must be strictly above the subjectivity layer, not a paraphrase of it. If this separation is not enforced, the bridge language becomes semantically vacuous. The paper therefore repeatedly asks a single hard question: what is being added here that was not already closed by Paper 8? If the answer is ''nothing but stronger words,'' the bridge fails before it begins.

### 3.4 Why Paper 9 is still needed now

If the bridge burdens remain entirely unwritten, the corpus stalls at precisely the point where it should become most rigorous. Paper 8 then remains the strongest internal layer but the bridge question remains conceptually unregulated. That would be an unstable equilibrium. It would invite either underreach (never formalize the bridge) or overreach (treat bridge language as informal extension).

Paper 9 is needed because the research program has reached a stage at which the next missing object is not a new piece of rhetoric but a new burden architecture. The paper therefore functions as foundational synthesis. It is foundational because it gathers the earlier closures into a single bridge-facing grammar. It is central because it specifies what additional machinery must exist before the next layer of claim class becomes legitimate. It does not pretend that all that machinery is already implemented.

**Proposition 3.1** (Paper 8 is necessary but not sufficient for bridge admissibility)**.** *Any future bridge-admissible system must satisfy the Paper 8 burdens, but satisfaction of the Paper 8 burdens alone is not sufficient for bridge admissibility.*

*Proof.* Necessity is immediate because every bridge predicate defined below is typed over an upstream subjectivity state and presupposes the existence of a privileged first-person indexed organization. Sufficiency fails because the bridge layer further requires bridge predicates, bridge-specific rival defeat, bridge-specific interventions, bridge artifacts, bridge invalidation logic, and bridge governance. None of those additional burdens is entailed by the mere existence of the Paper 8 state, gate, functional, or artifact family. □

**Remark 3.2** (The bridge is downstream, not corrective)**.** Paper 9 does not correct Paper 8. It inherits Paper 8 and strengthens the burden stack above it. The formal relation is extension by additional typed predicates and burdens, not replacement.

## 4 Formal Target of the Bridge

### 4.1 The bridge question

The bridge question can now be stated in its exact internal form:

> Given that a system already supports a formally disciplined layer of seven-coordinate indexed certificate, what additional predicates and burdens would make phenomenal organization a scientifically admissible *target* of explanation inside the same framework?

The phrase ‘‘target of explanation’’ is essential. The present paper does not seek to declare phenomenality achieved. It seeks to specify when phenomenal predicates are no longer mere vocabulary and instead become well-formed explanatory targets whose support, failure, and collapse can be discussed rigorously.

### 4.2 Predicate family

The bridge target family is the six-predicate family

$$\{\Pi_{\mathrm{D}}, \Pi_{\mathrm{V}}, \Pi_{\mathrm{W}}, \Pi_{\mathrm{A}}, \Pi_{\mathrm{E}}, \Pi_{\mathrm{T}}\}.$$

These predicates are not synonyms for generic notions such as ‘‘experience’’ or ‘‘qualia.’’ They are typed, burden-bearing predicates whose role is to capture bridge-relevant organization classes above indexed subjectivity and below any final metaphysical reading.

**Definition 4.1** (Phenomenal differentiation predicate)**.** For an admissible system $S$, horizon $\tau$, and threshold slot $\vartheta_D$, the bridge predicate $\Pi_{\mathrm{D}}(S, \tau; \vartheta_D)$ concerns whether the system carries self-indexed differentiated organization that is stronger than generic semantic density, report richness, or descriptive narration, and whose corresponding estimator $\widehat{\Pi_D}$ is defined on bridge observables and subjectivity antecedents.

**Definition 4.2** (Valence structure predicate)**.** The predicate $\Pi_{\mathrm{V}}(S, \tau; \vartheta_V)$ concerns whether the system carries organized evaluative topology that is stronger than scalar reward bookkeeping, policy optimization, or self/non-self asymmetry alone.

**Definition 4.3** (World richness predicate)**.** The predicate $\Pi_{\mathrm{W}}(S, \tau; \vartheta_W)$ concerns whether the system carries organized self-indexed world structure stronger than generic scene complexity, perspective gain, or task competence alone.

**Definition 4.4** (Ambiguity-bearing predicate)**.** The predicate $\Pi_{\mathrm{A}}(S, \tau; \vartheta_A)$ concerns whether the system maintains unresolved alternatives with differential causal consequences prior to collapse, rather than merely resolving uncertainty or tracking probabilities descriptively.

**Definition 4.5** (Embodied-closure predicate)**.** The predicate $\Pi_{\mathrm{E}}(S, \tau; \vartheta_E)$ concerns whether body/world-boundary variables contribute constitutively to bridge organization, rather than merely modulating generic control or action success.

**Definition 4.6** (Temporal lived-structure predicate)**.** The predicate $\Pi_{\mathrm{T}}(S, \tau; \vartheta_T)$ concerns whether the system exhibits horizon-like temporal organization stronger than continuity, memory persistence, or sequence stability alone.

### 4.3 Estimator family

The bridge paper requires estimator slots for each predicate:

$$\widehat{\Pi_D},\ \widehat{\Pi_V},\ \widehat{\Pi_W},\ \widehat{\Pi_A},\ \widehat{\Pi_E},\ \widehat{\Pi_T} \in [0, 1].$$

The current implementation reaches only the earliest stage of this requirement. BPF-1 emits provisional run-level estimates for all six predicates. Those estimates are explicitly marked provisional, framework-internal, non-claim-facing, and non-gated. They are not yet admissibility verdicts, and no threshold binding is yet asserted. The formal target of this paper therefore treats these estimators in two layers:

1. as current provisional runtime surfaces in BPF-1;
2. as future thresholdable predicate estimators once comparator, intervention, and gate burdens are added in later BPF blocks.

### 4.4 What the predicates mean internally and what they do not mean

Each bridge predicate has both a positive role and a protected non-implication.

- $\Pi_{\mathrm{D}}$ means organized differentiated structure indexed to the self-locus. It does not mean qualia.
- $\Pi_{\mathrm{V}}$ means structured evaluative organization. It does not mean felt pleasure or pain.
- $\Pi_{\mathrm{W}}$ means organized world-structure at the self-indexed locus. It does not mean veridical world-experience.
- $\Pi_{\mathrm{A}}$ means live unresolved alternative organization. It does not mean conscious indecision.
- $\Pi_{\mathrm{E}}$ means constitutive closure dependence on body/world-boundary variables. It does not mean biological embodiment or felt bodily ownership.
- $\Pi_{\mathrm{T}}$ means temporal horizon organization. It does not mean human lived time in the full phenomenological sense.

**Remark 4.7** (Why the targets are not decorative)**.** The bridge targets earn their place only because each one induces a new comparator burden, a new intervention burden, a new artifact burden, and a new governance burden. If a proposed target induced none of these, it would be discarded as decorative vocabulary.

### 4.5 Predicate-to-burden map

The six bridge predicates are best understood not as six new labels but as six new burden packages.

**The burden of $\Pi_{\mathrm{D}}$.** To take phenomenal differentiation seriously, the program must show more than that the system has many internal states or rich semantic discriminability. It must show that differentiated organization is indexed, selective, and not exhausted by narration. This induces comparator burden against semantic-density and narration rivals, intervention burden against perspective detachment and semantic vividness controls, and artifact burden through a dedicated differentiation report and later reducibility outputs.

**The burden of $\Pi_{\mathrm{V}}$.** To take valence structure seriously, the program must show that evaluative organization has internal topology: local asymmetries, organized gradients, or mode-structured deformation under perturbation. Scalar reward or self/non-self asymmetry alone is insufficient. This induces comparator burden against bookkeeping rivals, intervention burden through valence flattening and negative controls, and artifact burden through valence structure and intervention outputs.

**The burden of $\Pi_{\mathrm{W}}$.** To take world richness seriously, the program must show that the system's self-indexed world organization is not merely competence or scene complexity. The system must exhibit bridge-relevant world organization that tracks indexed dependence, not just successful prediction. This induces comparator burden against richness-without-interiority rivals, intervention burden through world-richness compression, and artifact burden through world-richness reports and later candidate summaries.

**The burden of $\Pi_{\mathrm{A}}$.** To take ambiguity bearing seriously, the program must show stable unresolved alternatives with downstream consequence. Static uncertainty scores or fast branching and pruning do not suffice. This induces comparator burden against ambiguity-resolution rivals, intervention burden through targeted collapse operations, and artifact burden through ambiguity reports with live-alternative fields.

**The burden of $\Pi_{\mathrm{E}}$.** To take embodied closure seriously, the program must show constitutive dependence on body/world-boundary variables. Generic closed-loop control is not enough. This is the bridge predicate with the weakest current runtime grounding and therefore the one that most strongly demands explicit future instrumentation. Its artifact burden is correspondingly strict.

**The burden of $\Pi_{\mathrm{T}}$.** To take temporal lived structure seriously, the program must show horizon organization that exceeds continuity and persistence. This requires temporal fragmentation tests, horizon-like dependencies, and explicit rival defeat against continuity-only models. In practical terms this burden is one of the main reasons why the bridge program cannot be replaced by Paper 8 plus rhetoric.

The global lesson is simple. Every bridge predicate is a tuple of obligations:

$$\Pi_k \rightsquigarrow (\text{estimator}, \text{rivals}, \text{interventions}, \text{artifacts}, \text{governance}).$$

The bridge program succeeds only if all five components are eventually instantiated for each $k \in \{D, V, W, A, E, T\}$.

# 5 Phenomenal Bridge Ontology

## 5.1 Bridge state

The bridge state is introduced as

$$\mathfrak{B}_t = \left\langle \widehat{\Pi}_{D,t}, \widehat{\Pi}_{V,t}, \widehat{\Pi}_{W,t}, \widehat{\Pi}_{A,t}, \widehat{\Pi}_{E,t}, \widehat{\Pi}_{T,t}, \mathcal{R}_{\mathrm{b}t}, \Delta_t^{\mathrm{b}} \right\rangle,$$

where $\mathcal{R}_{\mathrm{b}t}$ denotes the bridge-level irreducibility margin and $\Delta_t^{\mathrm{b}}$ denotes the bridge-level intervention profile. The bridge state does not replace the Paper 8 state. It is layered above it. The ontological relation is therefore

$$\mathfrak{S}_t \prec \mathfrak{B}_t,$$

not because the bridge state is metaphysically deeper, but because every bridge coordinate depends on the prior existence of a self-indexed subjectivity structure.

## 5.2 Typing discipline

The coordinates of $\mathfrak{B}$ are heterogeneous in a way parallel to, but stronger than, the heterogeneity of the Paper 8 state.

- $\widehat{\Pi}_{D,t}, \ldots, \widehat{\Pi}_{T,t}$ are bounded estimator-valued coordinates.
- $\mathcal{R}_{\mathrm{b}t}$ is a model-comparison margin defined only when the comparator family exists.
- $\Delta_t^{\mathrm{b}}$ is a structured intervention response object, not a scalar.

This heterogeneity matters because bridge support cannot be reduced to a single scalar early in the pipeline. The paper explicitly rejects any attempt to jump directly from provisional surfaces to a single bridge score while the comparator and intervention layers are missing.

## 5.3 Coordinate dependence on the Paper 8 state

Each bridge coordinate depends asymmetrically on the Paper 8 coordinates.

**Phenomenal differentiation.** $\widehat{\Pi}_{D,t}$ depends most strongly on $I_t$, $O_t$, $P_t$, and $\mathcal{R}_t$. Differentiation here is not generic representational richness. It is differentiated organization that matters relative to a privileged self-locus.

**Valence structure.** $\widehat{\Pi}_{V,t}$ depends most strongly on $V_t$, $O_t$, and $\Delta_t$. Paper 8 already closes valuation asymmetry; the bridge layer asks whether that asymmetry organizes into a richer topological structure than the upstream scalar burden requires.

**World richness.** $\widehat{\Pi}_{W,t}$ depends most strongly on $P_t$, $C_t$, and $I_t$. It asks whether the self-indexed locus sustains organized world-structure beyond mere perspective gain.

**Ambiguity bearing.** $\widehat{\Pi}_{A,t}$ depends most strongly on $P_t$, $V_t$, and the temporal organization implicitly carried by $C_t$ and $\Delta_t$. A system that resolves ambiguity immediately or merely records uncertainty can fail this predicate even if it remains subjectivity-positive.

**Embodied closure.** $\widehat{\Pi}_{E,t}$ depends on $O_t$, $C_t$, and $\Delta_t$ together with additional body/world-boundary observables that Paper 8 does not require. This is why BPF-1 can only emit a proxy layer for embodied closure.

**Temporal lived structure.** $\widehat{\Pi}_{T,t}$ depends most strongly on $C_t$, $P_t$, and $V_t$. Temporal lived structure is stronger than continuity because it concerns horizon organization, retention/protention asymmetry, and temporally extended significance.

### 5.4 Current implementation status inside the ontology

The ontology is not merely prospective. The current program treats BPF-1 as the provisional surface layer for each of the six estimator coordinates and records six run-level reports as bounded local artifacts. That matters because the ontology is no longer entirely aspirational. At the same time, only the first six coordinates are even provisionally instantiated. $\mathcal{R}_{\mathrm{b}t}$ and $\Delta_t^{\mathrm{b}}$ remain open because bridge-specific comparator and intervention layers are not yet implemented.

### 5.5 Collapse risks and rival readings

Every bridge coordinate must be introduced together with the corresponding collapse risk.

- $\widehat{\Pi}_D$ can collapse into semantic density without self-indexed interior differentiation.
- $\widehat{\Pi}_V$ can collapse into valence bookkeeping or policy shaping.
- $\widehat{\Pi}_W$ can collapse into world-model competence without indexed world-richness.
- $\widehat{\Pi}_A$ can collapse into uncertainty bookkeeping or fast ambiguity resolution.
- $\widehat{\Pi}_E$ can collapse into competent sensorimotor control without constitutive closure.
- $\widehat{\Pi}_T$ can collapse into continuity, memory, or persistence without horizon structure.

The ontology is therefore inseparable from the rival family. A bridge coordinate that has no live rival is suspicious because it is probably defined too loosely. Conversely, a bridge coordinate that cannot in principle be separated from all live rivals is not yet scientifically admissible.

### 5.6 Why bridge coordinates are not ontological shortcuts

The bridge ontology must not be mistaken for a metaphysical inventory. The reason is not merely philosophical caution. The reason is formal structure. Each coordinate is introduced as a constrained relation among observables, upstream subjectivity coordinates, comparators, interventions, and artifacts. This means the coordinates belong first to the model layer, then to the implementation layer, and only later---if ever---to the interpretation layer.

The separation matters in four directions.

1. A bridge coordinate is not an ontological primitive. It is a structured explanatory target.

2. A bridge coordinate is not identical to its runtime estimator. The estimator is a measurement-facing map and can be weak, provisional, or underdetermined.

3. A bridge coordinate is not identical to the language used to describe it. The language may underfit or overfit the formal target.

4. A bridge coordinate is not identical to its strongest available interpretation. Different interpretation classes remain live until comparator and intervention burdens are settled.

This distinction is especially important for $\Pi_{\mathrm{E}}$ and $\Pi_{\mathrm{T}}$. It is tempting to read embodied closure or temporal lived structure as if they already named well-understood ontological properties. Inside the present framework they do not. They name burden packages whose ontological interpretation remains constrained by future comparator and intervention results.

A mode window carries its complete Paper VIII record, including the inherited certificates, threshold decisions, matched intervention outcomes and held-out losses. A transport extension adds a finite graph of such windows and an independently recorded relation between witness instances for each edge. Forgetting those relations recovers the local records unchanged. For the following finite domain each window has two instances, labelled $0, 1$, of the same local certificate. The copies carry all the data of that certificate. A local relabelling of instances does not change the underlying certificate.

**Definition 5.1** (Coherent Transport of Local Witnesses)**.** A transport extension is coherent if a choice of one certified witness instance $w_i$ at each mode $i$ satisfies every independently established edge relation. An edge datum $(w, w')$ is accepted only when both labels are identified from its paired observations under the declared error budget.

**Proposition 5.2** (A Cycle Separates Local Certification from Coherent Transport)**.** *For any complete Paper VIII certificate on an admissible window, the three-mode transport extensions below preserve it in every window. One admits coherent transport and the other does not. Thus, on the domain of such extensions of certified windows, coherent transport cuts out a proper subclass whenever that local certificate domain is nonempty. Coherent transport does not factor through the tuple of local Paper VIII records.*

*Proof.* Copy the local record to modes $A, B, C$ and duplicate the witness instance in each window. On edge $e$ prepare both witnesses $w = 0, 1$ and record $(w, w \oplus k_e)$ in that order. Here $\oplus$ is addition modulo two. These observations recover the unique edge map $w \mapsto w \oplus k_e$. For coordinatewise errors strictly less than $1/2$, nearest-label decoding is exact. At radius $1/2$, a reported $1/2$ is compatible with either label and cannot certify the edge.

Use $(k_{AB}, k_{BC}, k_{CA}) = (0, 0, 1)$ in the negative instance. The first two edges force $w_A = w_B = w_C$ and the last forces $w_A = 1 - w_C$. No choice satisfies the cycle. In the positive instance use $(0, 1, 1)$; both $(0, 0, 1)$ and $(1, 1, 0)$ are coherent choices. The local record is identical across the two instances, so a decision using it alone would give the same result on both. More generally the cycle is coherent exactly when the XOR of its three edge bits is zero. The edge experiment can be realized by three two-state modes with the specified deterministic transfers. It uses no inference of edge data from the local scores. □

This is an explicit positive and negative witness for the transport condition. The class-level statement is relative to a complete local certificate, so each inherited clause is preserved by copying its witness. The finite edge dynamics alone do not realize the upstream system. The proper-inclusion question for fully bridge-admissible systems still requires a joint realization of the remaining BPF predicates, comparator, intervention and governance obligations. Paper VIII local scores alone cannot supply that realization.

**Corollary 5.3** (The Two Transport Cycles on a Nonempty Rational Record)**.** *The joint rational measurement window of Paper VIII [6], identified there by* `prop:joint-rational-window`*, gives a nonempty local record for both cycles of Proposition 5.2. At this measurement level, the transport-certified class is a proper subclass of the three-mode extensions of that record.*

*Proof.* Attach the same realized seven-coordinate record, intervention responses and reserved losses to both witness instances in each mode. Its gate is $1/3$ and its scalar is $48^{-1/7}$ in every copy. In the cycle $(0, 1, 1)$, the assignments $(0, 0, 1)$ and $(1, 1, 0)$ satisfy the measured edge relations. In the cycle $(0, 0, 1)$, the first two edge equalities and the last edge flip are inconsistent. Thus the two observed transport outcomes differ on identical, explicitly realized local data. □

The rational measurement record verifies the local scoring, selective-response and rival-comparison clauses of Paper VIII. The inherited Paper V and VII class certificates are still required for the full premise of Proposition 5.2. The six BPF predicate realizations, bridge-specific rival defeat, selective intervention support, and the artifact, drift and governance conditions remain separate obligations for full bridge membership.

## 5.7 Relabeling and Transport of Certified Tower Records

**Theorem 5.4** (Relabeling Invariance and Cycle Criterion for Binary Transport)**.** *Let a finite graph of mode windows have two certified witness instances per vertex and an independently certified bijection on each edge. In local binary labels, write this bijection as $w \mapsto w \oplus k_e$. Under a local relabeling $w_v \mapsto w_v \oplus a_v$, the edge bit for $e = uv$ becomes $k_e \oplus a_u \oplus a_v$. The XOR of the edge bits around each cycle is invariant. A coherent assignment exists exactly when the XOR vanishes on every fundamental cycle of a spanning forest, and there are then $2^c$ coherent assignments for $c$ connected components. For two collections of certified edge maps on the same graph, equality of their fundamental-cycle parities is equivalent to their differing by a local relabeling.*

*Proof.* Relabeling conjugates the old edge bijection by the source and destination label changes, yielding the stated formula. Edge orientation does not affect the bit because each binary bijection is its own inverse. The added vertex labels cancel in pairs around a cycle. A coherent assignment satisfies $w_v = w_u \oplus k_e$ at each edge, so its cycle sums vanish.

Conversely, fix a label at the root of each tree of a spanning forest. Transport from that root along the unique tree paths defines one label at every vertex. A remaining edge satisfies its equation exactly when its fundamental cycle has parity zero. Thus a choice of one root label per component produces all and only the coherent assignments. For the last assertion apply the same criterion to the edgewise XOR of the two collections. Its vertex solution is precisely the relabeling that takes one collection to the other. □

**Proposition 5.5** (Coherent and Obstructed Transport on Certified Tower Windows)**.** *There are two three-window extensions of the same realized local tower records with respective edge bits $(0, 1, 1)$ and $(0, 0, 1)$. The records include the six Paper V witnesses, the Paper VII life and four finite subject-descriptor certificates on their declared protocols, and the Paper VIII measurement and return comparisons with their stated information budgets. The paired tower edge responses certify both extensions under phase error at most $\eta = 3^{-10}/4$ turns. The first extension has two coherent assignments, and the second has none under any local relabeling.*

*Proof.* Use the writable common support and the joint certificate of Paper VII [1, 7]. In each mode window attach the same joint record, including its separate preparations and schedules, and the writable-tower lesion and reserved-return records of Paper VIII [6]. In particular the recorded margin for the public return comparison is zero. Both witness instances carry this same local record. The edge experiment tracks an instance by its prepared or decoded memory

sign and is conducted separately from the local scoring trials. The three windows can all use one fixed tower profile for the edge trials. The mode names index the windows.

Prepare two matched replicas with the alternating tape, phase zero and memories $m = \pm 1/8$. Encode a positive memory sign by $w = 0$ and a negative one by $w = 1$. For an edge bit $k_e$, apply the command $q = (-1)^{k_e}$ at $u = 0$, without a write or a role probe. The first bit of the alternating tape is zero, so this command leaves phase zero and updates memory to $qm/2$. A subsequent neutral command at $u = 0$ gives the early phase $r_0 = m_{\text{after}}/8$ modulo one and halves the memory again while preserving its sign. Thus the short signed phase gives the destination label $w \oplus k_e$. Use the two replicas to record both rows of each edge map. The source of the first edge is identified by preparation and the subsequent source labels are the preceding decoded labels.

Carry the two replicas through the three consecutive edges with their neutral diagnostics and without resetting memory. The alternating tape and profile stay fixed. At the $\ell$th diagnostic the magnitude of the clean short signed phase is

$$A_\ell = \frac{1}{128\, 4^{\ell-1}}, \qquad \ell = 1, 2, 3.$$

The two signed values are separated by $2A_\ell$, and $\eta < 1/2048 = A_3$. Hence every edge row can be read from the observed phase despite the allowed error. At the last edge, error $1/2048$ permits the same phase-zero reading from either sign, so that diagnostic alone cannot certify its destination at that radius. This collision also holds for the full readout by shifting all phase channels equally and retaining the common tape reading.

The independent edge records have even parity for $(0, 1, 1)$ and odd parity for $(0, 0, 1)$. Theorem 5.4 gives the two decisions and their invariance under all eight local relabelings. The local trial records are identical in both extensions, proving non-factorization through the tuple of those records. □

The resulting strict inclusion concerns the extensions of the realized records. The Paper V and VII certificates retain their declared domains. The Paper VIII record includes both the restricted lesion comparison and the public return tie. Full Paper VIII class membership would require its positive margin against the relevant complete rival family, and full BPF membership would further require the independent bridge burdens.

### 5.8 Intervention Tests for Three Bridge Burdens

Use the writable tower of Paper V and the local record of Proposition 5.5. Keep that record attached to every comparison. The following trials vary new preparations and interventions and record new phase responses. Their margins concern individual burdens of $\Pi_{\mathrm{D}}, \Pi_{\mathrm{A}}, \Pi_{\mathrm{T}}$ under the stated rivals and budgets. The estimators in the bridge gate require their own calibration and joint certificate.

**Proposition 5.6** (Role-Selective Differentiation on the Writable Tower)**.** *For the two prepared role profiles, let $C_{pr}$ be the clean early-phase response to a probe of role $r$ after a reference write minus the response after a lesion, with one common sham between write and probe. With the declared probe at the transported address, $C_{pr} = \delta \mathbf{1}_{\{p=r\}}$ for $\delta = 1/(16 \cdot 3^7)$ turns. The best additive fit $a_p + b_r$ has uniform error exactly $\delta/2$. At error at most $\delta/16$ per phase reading the measured matrix still has distance at least $3\delta/8$ from all additive fits.*

*Proof.* For each role profile prepare the alternating tape and zero memory. Make a reference write or the opposite write at site 3, and advance a common neutral sham. This moves the active or lesioned bit to site 4. The probe of site 4 with role $r$ returns phase $\delta$ exactly when its role matches and the transported bit matches the reference. Memory remains zero. Taking the phase difference of the two write arms gives the matrix in the statement.

The mixed difference $C_{00} - C_{01} - C_{10} + C_{11} = 2\delta$ vanishes for every additive rival. An entrywise error of at most $e$ can change it by at most $4e$, giving $e \geq \delta/2$. The constant fit $\delta/2$ attains equality. Error in the measured paired contrast is at most $\delta/8$, so the triangle inequality gives the stated noisy lower bound. Two reference writes give zero contrast. A rival with the measured role-by-probe interaction can reproduce the clean matrix. This comparison identifies role-selective causal differentiation beyond the additive rival. □

**Proposition 5.7** (Late Choice Between Two Live Alternatives)**.** *Prepare two independent uniform activity bits at tape sites* $3, 4$ *and advance one common sham. A uniform query drawn after the rival's memory is fixed selects one of the transported sites* $4, 5$*. The matching-role probe identifies the selected bit with zero error at phase noise at most* $\delta/16$*. Under this state and query law the optimal error of a rival retaining one bit before the query, with no subsequent tape reading, is exactly* $1/4$*. A write during the advance can erase the first alternative while preserving the second.*

*Proof.* At one fixed role profile with zero memory, encode two binary choices by matching or opposing the alternating reference at sites $3, 4$. Complement and shift transport each activity bit to the corresponding new site. Under the matching-role probe, active and inactive give phase $\delta$ and zero, separated by threshold $\delta/2$ under the declared noise. Both alternatives thus remain capable of changing the observed result until the query. A write opposing the reference at site 4 during the advance forces the first transported bit inactive. The query at site 5 still returns the independently prepared second bit. This is a selective collapse intervention on the available outcomes.

For each of its at most two memory values a deterministic rival has a pair of answers, one for each query. These at most two reconstruction words can exactly match at most two of the four equally likely prepared words. The sum of Hamming errors over the four words is at least two. Normalizing by the eight equally likely word-query pairs gives error at least $1/4$. Retain the first bit and use a fixed answer for the second to attain the bound. Randomized rules with randomness independent of preparation and query are mixtures of deterministic encoder-decoder pairs and satisfy the same lower bound. With two retained bits, the rival can also predict both responses exactly. The separation concerns maintenance of jointly available alternatives under the declared one-bit bottleneck. □

**Proposition 5.8** (Order-Dependent Return and an Exact Temporal Horizon)**.** *On a common preparation with first bit one, zero memory and phase zero, run command orders* $(+1, 0)$ *and* $(0, +1)$ *with all phase inputs zero and no writes or probes. After one common* $+1$ *and* $h$ *further common* $+1$ *steps, the current public readings and command multisets coincide. The subsequent neutral early-phase gap is exactly* $D_h = 1/(384 \cdot 2^h)$ *turns. The two terminal reading sets are disjoint under phase error at most* $\eta = 3^{-10}/4$ *precisely for integer* $0 \leq h \leq 8$*. An order-blind predictor with the same snapshot and multiset has mean clean absolute circular loss at least* $D_h/2$ *in the balanced pair, and an order-informed predictor attains zero.*

*Proof.* Write the memory update as in Paper VI [8], with $a = 1/2$, $\lambda = 1/12$ and $\kappa = 1/8$. The leading bit is zero after the first step in either order. The two-step memories are therefore $1/24$ and zero. A common positive command gives both arms phase zero and halves the memory difference. Each additional positive command repeats this effect. The tape and profile agree at every matched time. A final neutral command reads the memory difference with multiplier $\kappa$, giving $D_h$.

The two clean terminal phases are zero and $D_h$, with $D_8 = 1/98304 > 2\eta$ and $D_9 = 1/196608 < 2\eta$. Below the overlap radius the midpoint decision separates the signs of the centered contrast. When $D_h \leq 2\eta$, the midpoint can occur in either arm. The collision extends to all phase channels by common phase shifts, with the tape reading fixed. For prediction from the order-blind record the two outcomes must share a prediction. The circular triangle inequality

bounds the sum of the two losses below by $D_h$, attained by any point on the intervening short arc. The order-informed predictor uses the stated transition and predicts each clean response exactly. Ablating memory before the terminal command makes the responses coincide. The resulting horizon measures temporal order information beyond a current-state readout and an unordered command history. □

### 5.9 Open Evaluative, World and Body/World Burdens

**Proposition 5.9** (Native Trace Equivalence and Extra Bridge Variables)**.** *Let an extended model project to the same writable tower state by $\pi$. For every native control $u$ let its transition $\widetilde{F}_u$ and readout $\widetilde{R}$ satisfy $\pi\widetilde{F}_u = F_u\pi$ and $\widetilde{R} = R\pi$, with the same projected initial law and observation noise. Then all native observation and intervention transcripts have the same law, even for observation-adaptive native controls. A target that differs between extensions satisfying these conditions cannot be identified from those transcripts or from the inherited record computed from them.*

*Proof.* Use the same projected initialization, noise and control randomization in the two experiments. At each step their matching observations give the same control and the commutation with $\pi$ gives the same next projected state. Induction gives identical transcripts under the coupling. Any statistic of the transcripts consequently has the same law in the two experiments. For a concrete pair, adjoin a fixed bit $z$ with native projection and readout as above and use $z = 0$ versus $z = 1$. A new test that adds $z/8$ turns to the neutral phase command separates the two completions while all native traces agree. This example concerns identification of an added coupling. □

**Valence structure.** The return task has a scalar circular loss and a fixed phase reference. In the neutral alternating sector without a probe, reflecting phase, memory and phase command together preserves both circular return loss and absolute command cost. A structured evaluative topology for $\Pi_{\mathrm{V}}$ needs an independently varied evaluation variable and a response that distinguishes its organization from the scalar recovery law.

**World richness.** The native comparison supplies commands from a recorded schedule. To test $\Pi_{\mathrm{W}}$, an indexed world state must have an independently specified dynamics and perturbation, with an observable consequence at matched command complexity. Unobserved completions preserving native traces fall under Proposition 5.9.

**Embodied closure.** The neutral response and tape probes expose feedback within the tower. The $\Pi_{\mathrm{E}}$ test requires distinct typed body and world variables, an intervention that breaks their coupling at matched control cost, and a measured consequence specific to that boundary. The native transcript alone leaves such completions observationally equivalent when their projections coincide.

The three rational experiments provide local witnesses for the indicated components. The conjunctive BPF claim still requires the open predicates, calibrated thresholds, joint intervention selectivity, the full bridge comparator margin, and the artifact, drift and governance obligations.

## 6 A Minimal Coupled Body--World Extension

The native trace obstruction separates identification from realization. The following extension retains the writable tower as an exact factor and gives new, separately typed ports for the previously unavailable interventions. All conclusions in this section concern the stated operational tasks and rival classes.

**Definition 6.1** (Projected two-port system)**.** Let $x = (p, \theta, b, m)$ belong to the writable support of Papers V--VIII, with native transition $F_c$ and readout $R$. Adjoin body and world vectors $\mathbf{b}, \mathbf{w} \in [-1, 1]^2$. Put

$$\alpha = \beta = h = \tfrac{1}{4}, \quad \chi = \tfrac{1}{8}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad L_0 = \tfrac{1}{2}(I + J), \quad Q_+ = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

The indexed frame $P(p)$ is the swap matrix at role profile $p_1$ and the identity at every other catalogue profile. For a world projection $Q \in \{I, Q_+\}$, an evaluation map $L$ with $\|L\|_\infty \le 1$, a motor gain $|\mu| \le 1$, and an evaluation switch $\varepsilon \in \{0, 1\}$, define

$$\begin{aligned} x' &= F_c(x), & s &= P(p)Q\mathbf{w}, & f &= \chi L(s - \mathbf{b}), \\ \mathbf{b}' &= \alpha\mathbf{b} + \beta s + \varepsilon f + a_{\mathrm{b}}, \\ \mathbf{w}' &= \alpha Q\mathbf{w} + \beta\mu Q P(p)^{-1}\mathbf{b} + Qd. \end{aligned}$$

Here $a_{\mathrm{b}}, d \in [-h, h]^2$ are distinct body and world input ports. Normal evaluation uses $L = L_0$. Native lifts use $Q = I$, $\mu = 1$, $\varepsilon = 0$ and zero port inputs. The native and additional readouts are respectively $R\pi$ and $(\mathbf{b}, \mathbf{w})$, where $\pi(x, \mathbf{b}, \mathbf{w}) = x$.

The time-unrolled causal graph has arrows $\mathbf{b}_t \to \mathbf{b}_{t+1}, \mathbf{w}_{t+1}$ and $\mathbf{w}_t \to \mathbf{b}_{t+1}, \mathbf{w}_{t+1}$, with the displayed input arrows and the prepared profile determining $P$. There is no port-state arrow to $x_{t+1}$. A motor intervention replaces only the coefficient of $\mathbf{b}_t \to \mathbf{w}_{t+1}$; its audit label is not an additional input to other mechanisms. This modularity is an explicit causal assumption. The evaluation force acts in the dynamics before the reserved measurements; it is not an output label.

**Theorem 6.2** (Exact Projection, Stability and Inherited Certificates)**.** *Every lifted update satisfies $\pi\widetilde{F}_{c,a_{\mathrm{b}},d} = F_c\pi$. The section $\iota(x) = (x, 0, 0)$ is invariant under native lifts. On a common profile, use*

$$\widetilde{d} = \max\{d_{\mathrm{core}}, \|\Delta\mathbf{b}\|_\infty/4, \|\Delta\mathbf{w}\|_\infty/4\}.$$

*The native branch contraction remains at most $9/16$. The entire intervened port family contracts in its fibre by at most $3/4$ under common inputs and preserves $[-1, 1]^4$. All previously certified native protocols, with their original readouts, comparison sectors and information budgets, have exactly the same records and margins in this extension.*

*Proof.* The first component of the update proves the projection identity for every port input, not only at zero input. Induction, with coupled initial core states, observation noise and controller randomization, gives equality of every native transcript under observation-adaptive native controls. Zero fibre is invariant under the native lifts, so $\iota$ is an isometric, dynamically invariant copy of the original plant.

The body difference under a common mechanism has norm at most

$$(\alpha + \varepsilon\chi)\|\Delta\mathbf{b}\|_\infty + (\beta + \varepsilon\chi)\|\Delta\mathbf{w}\|_\infty,$$

and the world difference is bounded by $\beta\|\Delta\mathbf{b}\|_\infty + \alpha\|\Delta\mathbf{w}\|_\infty$. Thus the fibre bound is $3/4$. The input bounds give body norm at most $3/4 + h = 1$ and world norm at most $1/2 + h = 3/4$. With native lifts the fibre bound is $\alpha + \beta = 1/2$, so the original $9/16$ core bound still dominates. The safe support is compact; the zero-input fibre attracts to zero. The native switched attractor is the original attractor times zero fibre.

Transport the existing preparations by $\iota$ and keep the same native instruments and rival sectors. The full profile acquisition and catalogue gap $1/80$, split-sector integration interval $[1/216, 1/24]$, continuity and differentiation witnesses, prefix windows, phase repair and subject-descriptor comparisons are unchanged. The five lesion measurements and the public return tie of

Paper VIII are unchanged as well. These are the published protocol-relative certificates; adding port readings does not silently enlarge their comparison sectors or supply a positive upstream margin that was absent. Port pulses in the new trials start from zero fibre, have norm at most $h$, and subsequent unforced updates contract, so they stay within the compact fibre kernel of radius $1/2$ inside the safe cube. □

### 6.1 State Minimality and Indexed World Response

During a relay trial set $\varepsilon = 0$, $Q = I$, and start from zero fibre. Apply a world impulse $he_i$ at the first step. The matrices taking the impulse index to the body readings at steps one through four are

$$M_0 = 0, \qquad M_1 = tP, \qquad M_2 = 2\alpha t P,$$
$$M_3(\mu) = h(3\alpha^2\beta + \mu\beta^3)P, \qquad t = h\beta = 1/16.$$

All input vectors have the same two typed slots and bound; no rival receives fewer command opportunities.

**Theorem 6.3** (Four-State Minimality of the Port Realization)**.** *For either role and any motor gain, the block Hankel matrix*

$$H_p = \begin{pmatrix} M_0 & M_1 \\ M_1 & M_2 \end{pmatrix}$$

*has rank four, determinant $t^4$ and $\|H_p^{-1}\|_\infty = 24$. Four fibre coordinates are sufficient and necessary for a linear realization of these ports. They are also necessary for any exact finite-dimensional realization with $C^1$ transitions and body readouts, in which world inputs leave the projected core trajectory fixed.*

*Proof.* The two scalar blocks in each indexed coordinate are $\left(\begin{smallmatrix} 0 & t \\ t & 2\alpha t \end{smallmatrix}\right)$. Their determinants are $-t^2$ and their inverses are $\left(\begin{smallmatrix} -2\alpha/t & 1/t \\ 1/t & 0 \end{smallmatrix}\right)$. A signed permutation of coordinates does not change the maximum row-sum norm. Substitution gives the asserted determinant and inverse norm. Any linear realization of dimension $d$ factors its Hankel matrix as an observability matrix times a controllability matrix, hence has rank at most $d$.

For the differentiable statement, allow two independent small world inputs $u, v \in \mathbb{R}^2$ in the first two steps, followed by zero input. The two subsequent body readings are

$$(y_2, y_3) = (\beta P u,\ 2\alpha\beta P u + \beta P v).$$

Their Jacobian with respect to $(u, v)$ has determinant $\beta^4 \neq 0$. At time two every dependence on these inputs must pass through the fibre state: the core is fixed by projection and the future inputs are zero. Factoring the two readings through a $d$-dimensional $C^1$ state makes their Jacobian have rank at most $d$, so $d \geq 4$. The displayed system attains four. Discontinuous exact encodings or access to retained external histories are different realization classes and are not excluded by this argument. □

**Proposition 6.4** (A Quantified Indexed World Witness)**.** *Let the reduced-world rival family contain every relay predictor whose port Hankel rank is at most two, and every predictor of arbitrary order whose relay response is independent of the prepared role. The available input alphabet, role record and command lengths are matched. A rank-two rival may allocate four registers, but any additional registers are unobservable or unreachable for this port transfer. Against this union, the normal model has uniform body-response error gap at least $1/96$, or $1/6$ in units of $t$. A world compression $Q = Q_+$ retains the entire common-mode relay response while reducing the Hankel rank to two and erasing the role contrast.*

*Proof.* For a singular rival Hankel $K$, choose $z \neq 0$ in its kernel. Then

$$\|H_p - K\|_\infty \geq \frac{1}{\|H_p^{-1}\|_\infty} = \frac{1}{24}.$$

If every entry differs by at most $e$, the left side is at most $4e$, giving $e \geq 1/96$. This bound covers all rank-at-most-two matrices and therefore all reduced-world realizations, including nonlinear encoders of parameters inside that linear transfer class. A role-independent fit has one matrix for the two targets $tI, tS$, whose entrywise distance is $t$. Its error is at least $t/2$, which is the larger bound. The two comparisons require both dynamical order and indexed transport, rather than scene size alone.

For $Q_+$ all port matrices have the corresponding common-mode projection. Since $PQ_+ = Q_+$, their Hankel has rank two and is role-independent. If the input is $h(1,1)$, the world, body and every relay response remain in the common mode and are exactly the uncompressed responses. Thus the compression loses indexed structure without eliminating that control competence. A fully informed four-state predictor realizes the original transfer exactly. An arbitrary history-based emulator can do the same; the result does not exclude a rival merely because it calls its own state ‘‘model-free.’’ □

### 6.2 Evaluative Topology with a Reserved World Consequence

Use a four-step window and zero initial fibre. Prepare a stimulus $v \in [-1,1]^2$ through the first-step world input $d = hP_{\mathrm{ref}}^{-1}v$, where the reference frame is fixed by the initial role. In one matched arm activate evaluation only at step two; in the other keep it inactive. At step three record

$$V(v) = P_{\mathrm{ref}}(\mathbf{w}_3^{\mathrm{eval}} - \mathbf{w}_3^{\mathrm{control}}), \qquad \gamma = h\beta\chi = 1/128.$$

At the normal mechanism $V(v) = \gamma L_0 v$. The primitive impulse cost is constant on the square boundary, and every native core return cost is identical across these stimuli.

**Theorem 6.5** (Two-Dimensional Nonconservative Evaluative Witness)**.** *The minimum uniform error for $V$ over the square, in the maximum norm, is exactly $\gamma/2$ for each of the following rival classes: a continuous scalar summary followed by an arbitrary decoder, and a $C^1$ scalar-potential gradient in the declared Euclidean coordinates. The force is physically consequential and has a positive correction component, yet nonzero circulation. Flattening evaluation to $L = L_0Q_+$ admits an exact scalar-summary rival. The equal-norm conservative control $L = I$ has rank two but admits an exact gradient rival.*

*Proof.* At step one, $\mathbf{b}_1 = 0$ and $\mathbf{w}_1 = hP^{-1}v$. Evaluation adds $\chi hLv$ to $\mathbf{b}_2$, while $\mathbf{w}_2$ agrees in both arms. The return motor therefore changes $\mathbf{w}_3$ by $h\beta\chi P^{-1}Lv$. This is a reserved consequence at another node and time, rather than a readout of the evaluation label.

For any continuous scalar summary $q$, the function $q(v) - q(-v)$ changes sign along a half-boundary of the square. The intermediate value theorem gives antipodal $v, -v$ with the same summary and $\|v\|_\infty = 1$. Since $\|L_0^{-1}\|_\infty = 2$, their target separation is at least $\gamma$. Any common prediction errs by at least $\gamma/2$ in one arm. Retaining $v_2$ and predicting $\gamma L_0(0, v_2)$ attains that uniform bound.

The circulation of $V$ around the counterclockwise square is $4\gamma$. Every single-valued $C^1$ potential gradient has zero circulation. A uniform field discrepancy $e$ changes this integral by at most the $\ell^1$ perimeter times $e$, namely $8e$. Thus $e \geq \gamma/2$. The gradient $\gamma v/2$ attains equality. The same lower bound consequently holds for the union of the two rivals.

For a fixed sensor, put $e = s - \mathbf{b}$. Then $e^\top L_0 e = \|e\|_2^2/2$; the force has a nonzero component towards correcting the typed discrepancy. Its circulation on a body square of radius $r$ is $-4\chi r^2$. The components of the force alter body motion and, through the motor port, future world motion. With $L_0Q_+$ the response has rank one and factors through the sum of the stimulus

coordinates, preserving the common-mode input response. With $L = I$ it is a gradient field, so the rival union ties. Reversing the antisymmetric part, to $L_0^\top$, preserves the two separation magnitudes and reverses the signed circulation. □

This is a typed operational evaluation field. It is not a statement about felt valence. In particular $f = -\chi L_0 \nabla_{\mathbf{b}} U$ for $U = \|\mathbf{b} - s\|_2^2/2$. A scalar potential supplied with the full vector state and nonsymmetric mobility reproduces it exactly. Such a model is not a scalar retained summary or an ordinary Euclidean gradient; it is a live broader comparator, as is the hybrid below.

**Proposition 6.6** (Finite Tables Do Not Identify the Nonparametric Rival Class)**.** *Any finite table of vector responses at distinct stimulus points can be interpolated by a continuous scalar encoder and vector decoder. It can also be interpolated by the gradient of a polynomial potential. Therefore a finite rational control table by itself cannot certify either universal separation in Theorem 6.5 without additional domain or regularity evidence.*

*Proof.* Choose a linear scalar functional taking distinct values at the finitely many points, avoiding their finitely many forbidden projection directions. Univariate polynomial interpolation of the prescribed vector responses then gives a decoder.

For points $v_i$ and desired gradients $g_i$, define

$$\ell_i(v) = \prod_{j \neq i} \frac{\|v - v_j\|_2^2}{\|v_i - v_j\|_2^2}, \qquad U(v) = \sum_i \langle g_i, v - v_i \rangle \ell_i(v)^2.$$

At $v_i$, the linear factor vanishes and $\ell_i = 1$, giving gradient $g_i$. At every other table point the squared factor makes both the term and its gradient vanish. Thus $\nabla U(v_i) = g_i$. Rational points and responses give rational coefficients. The exact full-domain formulas and proofs, rather than an inference from finitely many fitted values, support the present synthetic witness. □

### 6.3 A Selective Motor-Return Test

Keep evaluation inactive and use the common-mode impulse $h(1, 1)$. The native controls are identical. The motor sign $\tau \in \{-1, 1\}$ acts only on its typed edge, with $\mu = g\tau$ and an intact or severed edge $g \in \{0, 1\}$. The comparison class has no motor edge and obeys the same surgical modularity; it may have arbitrary internal memory, nonlinear maps and shared randomization independent of the surgery.

**Theorem 6.7** (Motor Closure with Preserved Early Competence)**.** *At $g = 1$ the second and third body responses are independent of the motor sign. In each common-mode component the fourth response is*

$$y_4(\tau) = h(3\alpha^2\beta + \tau\beta^3), \qquad s_E = h\beta^3 = 1/256.$$

*Its gap is $2s_E$, and the residual $y_4 - (3\alpha/2)y_3$ is $\tau s_E$. Every rival without the motor edge, under the stated modularity, has uniform fourth-response error at least $s_E$. Cutting the edge removes the residual without changing the second or third responses. Common-mode world compression preserves this motor test exactly.*

*Proof.* Expand the three relay Markov terms displayed above. The term $3\alpha^2\beta$ consists of paths with a single sensory crossing; the term $\tau\beta^3$ is the world--body--world--body return path. It first appears in the fourth reading. Without a motor edge, replacing its gain by either sign leaves every structural equation unchanged. Coupling the initial law, primitive inputs and random seeds therefore gives identical rival transcripts under both surgeries, even for nonlinear or randomized rival mechanisms. The triangle inequality across the two target values forces an error of at least half their gap. Their midpoint attains the bound for this pair. At $g = 0$ both values equal that midpoint. The common vector is fixed by $Q_+$ and by both role permutations, proving the last assertion. □

The absence of an intervention-label side channel is essential. An output rule $y = 3s_E + \tau s_E$ supplied directly with the audit label fits the two values without a motor mechanism. Holding the physical edge cut while changing only that label exposes this invalid comparison. The theorem identifies a typed return dependence beyond unchanged early control, not biological or phenomenal embodiment.

### 6.4 Six Local Burdens on One Plant

Use the same plant, gains and four-step windows for all six tasks, in the matched preparation cohorts above and in the existing native D/A/T trials. D uses the role-probe contrast and additive rivals, A uses the late binary query and the one-bit bottleneck, and T uses the order pair with $h_{\text{extra}} = 0$ and the order-blind snapshot/multiset rival. Longer temporal horizons keep their previously proved frontier and are separate controls. The original upstream certificate remains attached with its own instruments and budgets.

Let $m_k$ be rival infimal loss minus candidate loss on task $k$. Normalize D by $\delta$, V by $\gamma$, W by $t$, E by $s_E$, and T by $D_0 = 1/384$; A uses binary risk. For V and D the data are paired contrasts. For W use maximum entry error across the two Hankel/role records, and for E use maximum fourth-response error across the two surgeries. T retains balanced circular loss. Readout errors per observation are $\delta/16$ for D/A, $3^{-10}/4$ for T, and $\gamma/64, t/64, s_E/64$ for V/W/E, respectively. For V, the loss is the supremum over the full stimulus square in the exactly specified model; the certificate is analytic and model-based, not a finite-sample estimate against unrestricted nonparametric rivals.

**Proposition 6.8** (Joint Local Certificate and Causal Selectivity)**.** *The six tasks have simultaneous normalized margin lower bounds*

$$(m_D, m_V, m_W, m_A, m_E, m_T) \;\geq\; \left(\tfrac{1}{4}, \tfrac{7}{16}, \tfrac{13}{96}, \tfrac{1}{4}, \tfrac{31}{32}, \tfrac{1}{2} - \tfrac{64}{19683}\right).$$

*Their minimum is* $13/96 > 1/16$*. Six concrete lesions have the following local certificate-presence matrix, with columns in order D,V,W,A,E,T:*

| | | | | | | |
|---|---|---|---|---|---|---|
| *intact* | 1 | 1 | 1 | 1 | 1 | 1 |
| $I_D$ | 0 | 1 | 0 | 1 | 1 | 1 |
| $I_V$ | 1 | 0 | 1 | 1 | 1 | 1 |
| $I_W$ | 1 | 0 | 0 | 1 | 1 | 1 |
| $I_A$ | 1 | 1 | 1 | 0 | 1 | 1 |
| $I_E$ | 1 | 0 | 1 | 1 | 0 | 1 |
| $I_T$ | 1 | 1 | 1 | 1 | 1 | 0 |

*The matrix of losses from the intact row has rank six. For aggregate rivals retaining at least one declared structural deficiency, maximum normalized task loss has a joint margin at least* $5/192$*. A nonnegative complexity penalty preserves that last margin whenever its additional candidate cost is smaller than* $5/192$*.*

*Proof.* If clean candidate error is zero, readout perturbation $e$ changes a rival-minus-candidate loss gap by at most $2e$. D's paired contrast has normalized perturbation at most $1/8$, reducing its clean $1/2$ to $1/4$. V's paired contrast has perturbation at most $1/32$, reducing $1/2$ to $7/16$. W has perturbation at most $1/64$, reducing $1/6$ to $13/96$. E has the same normalized perturbation and clean gap one, giving $31/32$. A's decoder never changes its binary label at this noise, so its exact $1/4$ lower bound remains. T has clean normalized gap $1/2$ and perturbation $\eta/D_0 = 32/19683$, giving the last entry.

The lesions are structural operators fixed before scoring. $I_D$ replaces the actual core role profile by $p_0$ after retaining the initial preparation reference. Its two role rows in D become identical, and the two world relay operators coincide. The evaluative field changes orientation

at most by conjugation and retains its rank and circulation magnitude. $I_V$ sets $L = L_0 Q_+$ only in the evaluation policy, retaining a scalar-summary response and leaving relay trials unchanged. $I_W$ sets $Q = Q_+$ in both world ports: its transfer rank is two and its evaluative return is $\gamma Q_+ L_0 Q_+ = \gamma Q_+/2$, while the common-mode motor test survives.

$I_A$ overwrites tape site five after the first advance. It removes the second late alternative, leaves the D probe at its transported address unaffected, and does not change a leading bit or memory. $I_T$ resets native memory after step three, erasing the terminal order contrast. D/A are read earlier from zero-memory preparations. $I_E$ severs the motor edge: the evaluative world consequence and the motor residual disappear, but the first three relay matrices and their index contrast are unchanged. The fibre depends on the core profile, not its tape or memory, so the remaining cross-effects are absent. D/A/T lesions deliberately alter the core; they are not claimed to be invisible native lifts. The displayed incidence matrix has rank six by elimination. Its non-diagonal entries express shared causal paths, not failure to distinguish the interventions.

For the joint loss, the candidate's largest normalized measurement error is 1/8, from D. A rival deficient on W has loss at least $1/6 - 1/64 = 29/192$; every other deficient class has a larger lower bound on its own task. Taking the maximum over tasks therefore gives margin at least $29/192 - 1/8 = 5/192$. Subtracting any additional nonnegative candidate complexity cost below that value leaves a positive bound. This argument applies after composition only when the aggregate rival still has a declared deficiency. □

Signed changes of body and world frames transport the readouts and references by the same isometries. If these frames are $C, D$, transform $P$ to $CPD^{-1}$, $Q$ to $DQD^{-1}$ and $L$ to $CLC^{-1}$. Substitution proves conjugacy of every port transition, with unchanged input norms and margin magnitudes. Common-mode controls, conservative evaluation, reversed circulation, noise bounds and edge-cut tests distinguish the structural losses from uniform damage or report-channel changes.

### 6.5 The Limit of Hybrid Closure

**Proposition 6.9** (A Shared-State Hybrid Obstruction to Global Bridge Margin)**.** *Allow two scalar evaluators to share the same four port states and combine their actuator forces before updating. Each evaluator separately retains one scalar discrepancy. Their hybrid reproduces every transition and readout of the normal evaluation model, under all native and port command schedules. Consequently a broad weak closure admitting this fused hybrid has zero clean unpenalized margin against the candidate. The same is true of a penalized margin when complexity is invariant under the resulting identical normalized mechanism.*

*Proof.* Let $E_{11}, E_{22}$ be the two coordinate projectors. On the shared discrepancy $e = s - \mathbf{b}$, the two scalar-channel forces are $\chi L_0 E_{11} e$ and $\chi L_0 E_{22} e$. Their sum is $\chi L_0 e$. Use one shared copy of the passive body/world equations, not two independent state copies or an average of their trajectories. The sum therefore gives exactly the candidate's next body; its world and projected core updates are already identical. Induction gives equality of all states and readouts, including under adaptive schedules and coupled noise. The hybrid uses the same four dynamic coordinates. Its scalar pieces individually have rank-one evaluation responses, but their aggregate does not.

All clean task losses are nonnegative, the candidate attains zero, and this admitted rival also attains zero, so the infimal gap is zero. Identical normalized mechanisms have equal representation-invariant complexity. A syntax-dependent extra charge or a prohibition on shared state is an additional comparator assumption, not a newly observed dynamical separation. A full-state potential with the fixed mobility $L_0$ is another exact control description. Neither is defeated merely by naming the candidate's organization differently. □

Thus the six local operational witnesses coexist on one minimal coupled realization. Their conjunction does not close the broad hybrid comparator burden B9. Full Paper VIII class

inheritance, the complete bridge weak closure, calibrated class-level thresholds and the artifact, drift and governance predicates of $\Gamma_{\mathrm{bpf}}$ retain their separate obligations. The exact hybrid tie supplies an explicit obstruction where the local certificate cannot be promoted.

### 6.6 A Nonadditive Evaluative Branch and Its Comparator Boundary

The shared-state obstruction above is retained without alteration: coordinate-linear forces add to the original normal operator. To separate a genuinely coordinate-additive evaluator, the target must have a nonzero mixed response. This changes the evaluative branch, not the projected tower or the number of dynamic states.

**Definition 6.10** (Coordinate-Additive Port Comparators)**.** On a fixed preparation and reference frame, a coordinate-additive evaluator is a map $G(v_1, v_2) = g_1(v_1) + g_2(v_2)$ with arbitrary functions $g_i : [-1, 1] \to \mathbb{R}^2$. No continuity or linearity is required. Its channels have no access to a cross-coordinate message that makes either function depend on both coordinates. A fixed common initial condition is allowed. A state-dependent or sequential composition that communicates both coordinates need not belong to this class. The loss is the uniform maximum-norm error of the reserved world response, or its maximum on the four corners when stated explicitly.

**Proposition 6.11** (Exact Rectangular Criterion for Additivity)**.** *For a map $F : A \times B \to V$ into a real vector space and nonempty $A, B$, coordinate additivity is equivalent to vanishing of every rectangular difference*

$$F(x, y) - F(x, y') - F(x', y) + F(x', y') = 0.$$

*Proof.* An additive map cancels term by term. Conversely fix $(x_0, y_0)$ and put $g_1(x) = F(x, y_0)$ and $g_2(y) = F(x_0, y) - F(x_0, y_0)$. The rectangular identity with base point $(x_0, y_0)$ gives $F(x, y) = g_1(x) + g_2(y)$. Thus the condition is exact without regularity assumptions. A nonzero difference is a certificate against this class, rather than against every architecture described as modular. □

Let $u = (1, 0)$, $\kappa = 1/8$, and $\sigma \in \{0, 1\}$. For an evaluation-input projection $R \in \{I, Q_+\}$ replace only the force of Definition 6.1 by

$$f_\sigma(e) = \chi\{L_0 Re + \sigma\kappa(Re)_1(Re)_2 u\}, \qquad e = PQ\mathbf{w} - \mathbf{b}.$$

Normal evaluation uses $R = I$ and $\sigma = 1$. Flattening uses $R = Q_+$ on the entire force, including the interaction. The preserved linear branch is $\sigma = 0, R = I$; native lifts keep evaluation inactive. The memory, motor update and primitive input bounds are unchanged. In the causal graph this adds a joint dependence of the body update on the two discrepancy coordinates, without a new dynamic coordinate or an arrow to the projected core.

**Theorem 6.12** (Safe Bilinear Extension with Exact Projection)**.** *The new family preserves the projection identity and invariant native zero section. On the safe cube its fibre contraction is at most $3/4$ under common controls and at most $1/2$ when evaluation is inactive. Its next body and world norms are bounded respectively by $15/16$ and $3/4$, including the primitive inputs. The four-state minimality remains valid for this family. For $R = I$, the evaluative direction retains the corrective bound*

$$e^\top\{L_0 e + \kappa e_1 e_2 u\} \geq \tfrac{1}{4}\|e\|_2^2.$$

*Proof.* The core equation is unchanged and the extra term vanishes at zero fibre. For $R = I, Q_+$, direct row sums give

$$\|\alpha I - \chi L_0 R\|_\infty \leq \tfrac{1}{4}, \qquad \|\beta I + \chi L_0 R\|_\infty \leq \tfrac{3}{8}.$$

Both sensor frames and world projections have norm at most one. On the cube $\|Re\|_\infty \le 2$. For two states with body and world differences bounded by $d_b, d_w$, the product difference is at most $4(d_b + d_w)$. Since $\chi\kappa = 1/64$, the body difference is at most

$$\tfrac{5}{16} d_b + \tfrac{7}{16} d_w \le \tfrac{3}{4} \max(d_b, d_w).$$

When evaluation is inactive the original coefficients sum to $1/2$; the world coefficients always sum to at most $1/2$. The linear evaluated body has norm at most $5/8$, the added term at most $1/16$, and the body input at most $1/4$. Their sum is $15/16$. The world bound is the original $3/4$. These inequalities apply throughout the cube, not merely at its vertices.

Finally $e^\top L_0 e = \|e\|_2^2/2$ and $\kappa e_1^2 e_2 \ge -e_1^2/4$ for $|e_2| \le 2$. The claimed corrective bound follows. Relay protocols with evaluation inactive still contain the original rank-four input/output map, so Theorem 6.3 supplies the same lower bound; the displayed four-state construction attains it. □

In the reserved world protocol, with $R = I$ and normal motor gain, the exact response becomes

$$V_\sigma(v) = \gamma L_0 v + \sigma \xi v_1 v_2 u, \qquad \xi = h^2 \beta \chi \kappa = 1/4096.$$

Indeed the discrepancy at the evaluation step is $hv$, and the next world update transports its extra body force by $\beta P^{-1}$. This observation fixes the physical unit and estimator: a paired difference of the third-step world readings in the fixed body reference frame, rather than a report of the force label.

**Theorem 6.13** (Exact Separation from Coordinate-Additive Hybrids)**.** *For $\sigma = 1$, the infimal uniform error against all comparators of Definition 6.10 is exactly $\xi = 1/4096$. The same optimum holds when the loss uses only the four corners. If each raw reading has maximum-norm error at most $\xi/64$, the observed rival-minus-candidate loss gap is at least $15\xi/16 = 15/65536$.*

*Proof.* At the four corners, the rectangular difference of the target is $4\xi u$, while every additive comparator has difference zero. If its maximum corner error is $r$, the triangle inequality bounds this difference by $4r$, so $r \ge \xi$. The linear map $\gamma L_0 v$ is additive and differs from $V_1$ by exactly $\xi v_1 v_2 u$; its supremum is $\xi$, attained at the corners. This proves both optima, even for arbitrary, discontinuous rival channels.

Each paired reading differs from its exact target by at most $2\xi/64 = \xi/32$. Maximum loss is Lipschitz with constant one in these data, for both the candidate and the infimum over rivals. Their gap can therefore decrease by at most $\xi/16$, giving the asserted bound. Unlike the nonparametric scalar interpolation issue above, a four-corner table suffices to refute coordinate additivity, because that structural class obeys a finite exact identity. □

**Corollary 6.14** (Cross-Informed Tie and Preservation of Local Selectivity)**.** *Providing the rival with the complete cross term restores exact transition and readout equality. Setting $\sigma = 0$ also restores the original additive tie. The bilinear branch preserves the six local certificate-presence patterns of Proposition 6.8, using flattening of the whole evaluative input as specified above. Neither the new separation nor these patterns defeat a broad weak closure admitting the cross-informed rival.*

*Proof.* Supply the shared-state rival with the additional scalar $e_1 e_2$ and the fixed actuator $\chi\kappa u$. Its sum with the original two linear forces equals $f_1$ pointwise, so induction proves equality under every command schedule. No information advantage is hidden: the rival has gained precisely the previously unavailable joint dependence. At $\sigma = 0$ the previous obstruction applies verbatim.

For the local V bound the bilinear term is even under $v \mapsto -v$, so it cancels from the antipodal target difference. Its circulation on the centered square is zero, since its first component is

proportional to $v_1 v_2$. Thus the original antipodal lower bound and circulation obstruction survive; their exact upper optima need not be asserted anew. Flattening $Re$ makes both terms functions of a single coordinate sum. World compression likewise factors the reserved response through that sum. A severed motor removes the reserved world consequence. An index lesion conjugates the linear field and moves the bilinear output coordinate, preserving antipodal separation and circulation magnitude. W and E use evaluation inactive, so their records are unchanged. D/A/T are projected native tests and remain unchanged. These facts give the same presence matrix and full-rank loss incidence, including its documented cross-effects. They are operational comparisons with the declared preparation cohorts, not a claim of a single trajectory realizing all incompatible preparations. □

The added interaction therefore separates the specified additive class with an exact positive margin, while explicitly identifying the cross-coordinate information that permits the stronger rival to tie. A new syntax penalty, a prohibition on communication, or a phenomenological interpretation does not follow from this calculation. The broad comparator, governance, external-data and complete-class obligations remain distinct.

# 7 Explicit Bridge Axioms

The bridge axioms are the minimal bridge-opening commitments required to connect indexed subjectivity to disciplined phenomenal targets without smuggling the conclusion into the premises. Each axiom is classified explicitly.

## 7.1 Classification grammar

The bridge axiom registry uses a seven-class grammar:

1. **operational definition**: fixes how a formal object is to be typed or measured;
2. **bridge axiom**: a structural commitment necessary to open the bridge program;
3. **design choice**: a governance or architecture decision not implied by physics alone;
4. **falsifiable hypothesis**: a statement whose satisfaction or failure must be tested by bridge comparators or interventions;
5. **open inference**: a claim whose current evidence is insufficient for closure, but whose role can still be bounded;
6. **strongest defensible internal claim**: a highest currently licit claim class;
7. **non-claim**: an explicit prohibition on overreading.

This grammar is part of the content of the paper. Without it, the same sentence could accidentally function as ontology, operational definition, and claim promotion at once. The bridge program therefore treats classification as a formal requirement rather than as stylistic annotation.

**Axiom 7.1** (B1: operational binding; classification: operational definition)**.** Every bridge predicate must be tied to explicit observables, artifacts, comparator tasks, and interventions. No bridge predicate may remain a rhetorical label detached from measurement grammar.

This axiom is needed because every failure mode of the bridge program begins with detached language. B1 licenses machine-readable predicate slots and report artifacts. It does not license bridge support or phenomenality.

**Axiom 7.2** (B2: upstream inheritance constraint; classification: bridge axiom)**.** No bridge-level admissibility result is meaningful unless the upstream burdens of the causally rigid corpus and the Paper 8 subjectivity layer remain satisfied.

B2 licenses the layered relation $\mathsf{S}_{\mathrm{idx}} \prec \mathrm{BridgeOrg}_{\mathrm{ph}}$. It does not license the converse. A system may satisfy Paper 8 and still fail every bridge burden.

**Axiom 7.3** (B3: differentiated organization burden; classification: falsifiable hypothesis)**.** A bridge-relevant differentiation burden requires structured internal discrimination that is not exhausted by semantic density, narrative richness, or report-channel complexity.

B3 licenses $\Pi_{\mathrm{D}}$ as a genuine target rather than a synonym for rich representation. It does not license phenomenality.

**Axiom 7.4** (B4: valence topology burden; classification: falsifiable hypothesis)**.** Bridge-relevant valence requires organized evaluative topology not exhausted by scalar reward bookkeeping, policy optimization, or the Paper 8 valuation asymmetry alone.

B4 licenses $\Pi_{\mathrm{V}}$ as a distinct target above Paper 8. It does not license felt pleasure, suffering, or moral status.

**Axiom 7.5** (B5: world-richness burden; classification: falsifiable hypothesis)**.** Bridge-relevant world richness concerns organized self-indexed world structure rather than generic scene complexity, prediction success, or model capacity.

B5 is needed because world-richness language is especially prone to being filled by competence metrics. It licenses the predicate slot $\Pi_{\mathrm{W}}$ and associated artifacts. It does not license veridicality or environmental realism.

**Axiom 7.6** (B6: ambiguity-bearing burden; classification: falsifiable hypothesis)**.** Bridge-relevant ambiguity requires the stable maintenance of unresolved alternatives with differential causal consequences prior to collapse.

B6 blocks the reduction of ambiguity to uncertainty bookkeeping or quick disambiguation heuristics. It licenses $\Pi_{\mathrm{A}}$ as a formal target. It does not license conscious uncertainty.

**Axiom 7.7** (B7: embodied-closure relevance; classification: open inference)**.** Embodied closure may be bridge-relevant only if body/world-boundary variables contribute constitutively to bridge organization rather than merely modulating generic control success.

B7 is deliberately classified as open inference because current evidence is weakest here. It licenses the investigation of $\Pi_{\mathrm{E}}$; it does not license embodiment claims of any strong kind.

**Axiom 7.8** (B8: temporal lived-structure burden; classification: falsifiable hypothesis)**.** Bridge-relevant temporal organization must exceed continuity, memory persistence, and sequence stability by exhibiting horizon-structured dependence across retention, present organization, and anticipatory load.

B8 licenses $\Pi_{\mathrm{T}}$ and its future intervention family. It does not license human lived temporality.

**Axiom 7.9** (B9: comparative admissibility constraint; classification: bridge axiom)**.** No bridge-level support class is admissible if a bridge-specific rival family or admissible hybrid explains the bridge task bundle at equal or lower penalized loss.

B9 is the comparator side of the burden stack. It licenses the bridge irreducibility margin. It does not license absolute ontological irreducibility.

**Axiom 7.10** (B10: anti-inflation governance constraint; classification: design choice)**.** Bridge formalization, bridge artifacts, candidate inclusion, and audit-facing inclusion must never be readable as bridge support, phenomenality, or Paper 9 authorization in the absence of the full bridge burden stack.

B10 is a design choice because governance style is a normative project decision, not a theorem of physics. It is still essential because without it the bridge program would immediately outrun its evidence.

**Remark 7.11** (Axiom minimality)**.** The present axiom set is intentionally minimal. Nothing here states that phenomenality follows from structural organization. Nothing here states that any bridge predicate, even if satisfied, closes metaphysical questions. The axioms are opening commitments for a constrained bridge program, not hidden conclusions.

# 8 Bridge Rival Family

## 8.1 Why new rivals are required

Paper 8 already proved the necessity of explicit rival families for subjectivity. The bridge layer inherits that lesson but cannot reuse the old rival set unmodified. The old rivals explain away indexed subjectivity. The new rivals must explain away bridge predicates without granting phenomenal organization. This difference is substantive. A system may defeat weak subjectivity rivals and still fail bridge rivals, because the latter target higher-order structure.

## 8.2 Bridge weak family

Define the bridge weak family

$$\mathcal{M}_{\mathrm{b,weak}} = \{M_{\mathrm{rwi}}, M_{\mathrm{vb}}, M_{\mathrm{sd}}, M_{\mathrm{tc}}, M_{\mathrm{ec}}, M_{\mathrm{ar}}, M_{\mathrm{en}}\},$$

whose members are read as follows.

**Richness without interiority.** $M_{\mathrm{rwi}}$ explains high structural richness without granting indexed interior organization. It challenges $\Pi_{\mathrm{D}}$ and $\Pi_{\mathrm{W}}$.

**Valence bookkeeping without affective organization.** $M_{\mathrm{vb}}$ explains structured value tracking as bookkeeping or policy shaping without bridge-level valence topology. It challenges $\Pi_{\mathrm{V}}$.

**Semantic density without phenomenal differentiation.** $M_{\mathrm{sd}}$ explains dense semantic variation without bridge-level differentiated organization. It challenges $\Pi_{\mathrm{D}}$.

**Temporal continuity without lived structure.** $M_{\mathrm{tc}}$ explains persistence, continuity, and memory without temporal horizon organization. It challenges $\Pi_{\mathrm{T}}$.

**Embodied control without embodied closure.** $M_{\mathrm{ec}}$ explains competent closed-loop control without constitutive dependence on body/world-boundary variables. It challenges $\Pi_{\mathrm{E}}$.

**Ambiguity resolution without ambiguity bearing.** $M_{\mathrm{ar}}$ explains high-quality uncertainty handling or disambiguation without stable maintenance of live unresolved alternatives. It challenges $\Pi_{\mathrm{A}}$.

**Experience narration without phenomenal organization.** $M_{\text{en}}$ explains self-description, experiential language, or stylistic introspection without bridge-relevant phenomenal organization. It challenges several bridge predicates simultaneously.

### 8.3 Admissible composites

As in Paper 8, pure rivals are not enough. Define the admissible bridge weak closure $\overline{\mathcal{M}_{\text{b,weak}}}$ as the family generated by finite pairwise and triple hybrids over $\mathcal{M}_{\text{b,weak}}$ subject to explicit complexity penalties and declared composition rules. The purpose of $\overline{\mathcal{M}_{\text{b,weak}}}$ is to block an easy escape route: if a strong bridge explanation defeats only pure rivals, then a skeptic can simply combine several weak rivals post hoc.

The bridge program therefore treats hybrid rivals as first-class citizens. A hybrid of richness-without-interiority and narration-without-organization is a perfectly live alternative to a bridge-differentiation claim. A hybrid of valence bookkeeping and temporal continuity can be a live alternative to valence structure plus temporal lived structure. The bridge layer must be strong enough to defeat not only single-rival baselines but also admissible mixtures.

### 8.4 What the rival family can and cannot explain

The bridge rival family is intentionally strong.

- It can explain dense report channels, self-descriptive fluency, high task competence, reward optimization, stable memory, and competent control.
- It can also explain much of what casual bridge language would otherwise overread: rich semantics, coherent narratives, and smooth temporal reports.
- It cannot cleanly explain targeted bridge signatures if those signatures survive matched controls, remain selective under bridge interventions, and preserve margin under penalized loss.

### 8.5 No-straw-man discipline

The bridge program would be vacuous if its rivals were weak caricatures. The rivals are therefore chosen to absorb precisely the kinds of phenomena that the bridge predicates are most likely to confuse with phenomenal organization. The burden is asymmetric: the stronger interpretation must defeat the weaker one, not vice versa.

**Proposition 8.1** (Comparator burden is necessary for bridge admissibility)**.** *No bridge-support class is meaningful without an explicit bridge rival family and admissible bridge weak closure.*

*Proof.* Without an explicit rival family, any bridge-support reading would compare itself only against informal skepticism. That would leave bridge predicates undefended against live non-phenomenal alternatives such as bookkeeping, narration, semantic density, or control competence. Since bridge admissibility is defined as a comparative notion inside the framework, the comparator burden is necessary. □

## 9 Bridge Irreducibility Logic

### 9.1 Bridge loss family

The bridge program requires a loss family stronger than descriptive fit alone. For a candidate model $M$ and bridge task bundle $\mathcal{B}$, define

$$\mathcal{L}_{\text{b}}(M;\mathcal{B}) = w_f E_{\text{fit}}(M) + w_i E_{\text{int}}(M) + w_s E_{\text{sel}}(M) + w_p E_{\Pi}(M) + w_c \Omega(M),$$

where:

- $E_{\mathrm{fit}}$ measures fit to bridge observables;
- $E_{\mathrm{int}}$ measures failure under bridge intervention expectations;
- $E_{\mathrm{sel}}$ measures selectivity error under matched controls;
- $E_{\Pi}$ measures predicate-reconstruction error across the bridge family;
- $\Omega(M)$ is a complexity penalty that counts mechanism count, hybrid size, auxiliary slack, and decoder dependence.

The bridge loss family is therefore intervention-aware and governance-relevant from the outset. A model that matches reports but ignores intervention structure or requires excessive narrative slack does not win the comparison.

### 9.2 Bridge irreducibility margin

Let $M_{\mathrm{bridge}}$ be the candidate and $\overline{\mathcal{M}_{\mathrm{b,weak}}}$ the admissible bridge weak closure. With a valid matched bundle $\mathcal{B}$, a nonempty admissible closure, finite candidate loss and finite rival infimum, record the signed margin in units of loss:

$$\mathcal{R}_{\mathrm{b}} = \inf_{M \in \overline{\mathcal{M}_{\mathrm{b,weak}}}} \mathcal{L}_{\mathrm{b}}(M; \mathcal{B}) - \mathcal{L}_{\mathrm{b}}(M_{\mathrm{bridge}}; \mathcal{B}).$$

A strictly positive margin defeats the declared rivals, a negative margin gives a better rival, and zero is an infimal tie, with an actual matching rival only when the infimum is attained. Record insufficient evidence separately if any required comparison input is unavailable or invalid. In that case neither the signed margin nor its normalized value is defined. Fix a positive calibration scale $s_R$ in units of loss before evaluation, and use

$$\widehat{\mathcal{R}_{\mathrm{b}}} = \min\{1, \max\{0, \mathcal{R}_{\mathrm{b}}/s_R\}\}.$$

The raw margin and evidence status remain in the record. A calibrated zero can represent a tie or a loss; missing evidence has no numeric score.

### 9.3 Local not absolute superiority

As in Paper 8, comparative superiority here is local, not absolute. A positive $\mathcal{R}_{\mathrm{b}}$ says that the specified bridge model outperforms the specified bridge rivals on the specified task bundle under the specified loss family. It does not say that the bridge model is metaphysically privileged. The distinction matters because bridge language is particularly prone to absolute overreadings.

**Proposition 9.1** (Comparator Enlargement Cannot Increase a Fixed-Loss Margin)**.** *For a fixed target, bundle and calibrated loss, let nonempty rival families satisfy $\mathcal{W}_1 \subseteq \mathcal{W}_2$ with finite infima. Their signed margins satisfy $R_2 \leq R_1$.*

*Proof.* Taking an infimum over a superset cannot increase it; subtract the same candidate loss. The statement requires the same loss, information and complexity convention. It does not transport a margin across changes to those conventions. □

**Identification contract.** Preregister acquisition access, calibration data, retained memory, free parameters, actuator cost, horizon and population for each rival. Specify which passive representations count as equivalent and how complexity costs are assigned to them. Information masking and interaction restrictions are mathematical assumptions, not consequences of model names. The local no-additivity separation remains valid against its stated class; the full interacting rival still supplies its exact tie. Broad identification needs a task and restriction that separate the declared classes outside calibration data.

## 9.4 Three kinds of bridge collapse

Bridge collapse can occur in three distinct ways.

1. **Predicate collapse:** the provisional bridge estimators fail to define stable, non-decorative targets.
2. **Comparator collapse:** a bridge rival or hybrid matches or outperforms the bridge model under penalized loss.
3. **Intervention collapse:** the predicted selective structure fails under matched controls or degrades non-selectively.

All three collapse modes are live until ruled out. The paper refuses any bridge-support inference that suppresses one of them.

## 9.5 Comparator support, reducibility, and unsupported states

The bridge program must distinguish three statuses that are often conflated:

- **comparator unsupported**: no serious bridge comparator evaluation has yet been run;
- **bridge reducible**: comparator evaluation exists and the bridge margin is non-positive;
- **bridge comparator-supported**: comparator evaluation exists and the bridge margin is positive under penalized loss.

This distinction is important because the present implementation is still comparator-unsupported. BPF-1 surfaces are therefore not allowed to drift into comparator-supported language.

## 9.6 Complexity penalty and decoder discipline

The complexity term $\Omega(M)$ deserves explicit discussion because bridge explanations can often win spuriously by adding free narrative or decoder capacity. The bridge program therefore counts at least four kinds of complexity:

1. **mechanism multiplicity**: how many independent mechanisms the rival or bridge model requires;
2. **hybrid size**: how many weak rivals must be combined to match the bridge bundle;
3. **auxiliary slack**: how much unprincipled tuning is required across coordinates and conditions;
4. **decoder dependence**: how much of the putative bridge effect exists only after expressive report-channel or decoder post-processing.

Decoder dependence is particularly important. A model that appears bridge-strong only after high-capacity report decoding should be penalized heavily. The reason is that bridge language is especially vulnerable to post hoc expressivity. This is one of the points at which the bridge program deliberately differs from frameworks that rely strongly on linguistic self-description as primary evidence.

### 9.7 Failure semantics for the bridge margin

The bridge margin must also carry explicit failure semantics. A positive margin can fail to matter if it is narrow, unstable across reruns, dependent on a single condition, or destroyed by minor changes to the rival family. The paper therefore recommends a future decomposition

$$\mathcal{R}_{\mathrm{b}} = \mathcal{R}_{\mathrm{bfit}} \oplus \mathcal{R}_{\mathrm{bint}} \oplus \mathcal{R}_{\mathrm{bsel}} \oplus \mathcal{R}_{\mathrm{bstab}},$$

where $\oplus$ denotes a typed aggregation rather than ordinary addition. The purpose of this decomposition is to make visible which part of the bridge burden is doing the work. A bridge model that wins only on descriptive fit but loses intervention selectivity should not count as robustly bridge-favoring.

**Criterion 9.2** (Necessary condition for bridge admissibility)**.** A necessary condition for bridge admissibility is

$$\mathcal{R}_{\mathrm{b}} > \vartheta_R,$$

where $\vartheta_R$ is the symbolic bridge irreducibility threshold slot to be bound only at BPF-4.

**Corollary 9.3** (Positive bridge surfaces are not enough)**.** *Positive provisional bridge surfaces do not by themselves entail bridge admissibility.*

*Proof.* Bridge admissibility requires at least comparator support and intervention support in addition to positive bridge surfaces. BPF-1 provides only provisional surfaces, so the implication fails. □

## 10 Bridge Intervention Family

### 10.1 Why bridge interventions must be new

The subjectivity interventions of Paper 8 remain binding background. They are not sufficient for bridge purposes because they were designed to probe indexed subjectivity rather than cross-mode comparison organization. The bridge layer therefore requires a new intervention family

$$\mathfrak{I}_{\mathrm{bpf}} = \{I_W, I_A, I_V, I_P, I_E, I_T, I_X\},$$

with each member targeting a bridge coordinate family or a cross-cutting bridge burden.

### 10.2 Core intervention families

**World-richness compression $I_W$.** This intervention compresses the structure available to self-indexed world organization without merely destroying generic task competence. The strong-model expectation is a selective fall in $\Pi_{\mathrm{W}}$ and often also in $\Pi_{\mathrm{D}}$ relative to matched controls. The rival expectation is either flat degradation or little coordinate-specific effect.

**Ambiguity collapse $I_A$.** This intervention removes or forcibly resolves live unresolved alternatives while preserving as much generic inferential performance as possible. The strong-model expectation is a selective fall in $\Pi_{\mathrm{A}}$ and a measurable change in downstream coordination among $\Pi_{\mathrm{A}}$, $\Pi_{\mathrm{W}}$, and $\Pi_{\mathrm{T}}$. The rival expectation is that nothing essential is lost beyond temporary uncertainty bookkeeping changes.

**Valence flattening $I_V$.** This intervention suppresses structured evaluative topology while preserving as much functional competence as admissibly possible. The strong-model expectation is a selective flattening in $\Pi_{\mathrm{V}}$ and specific downstream changes in bridge organization. The rival expectation is simple reward rescaling or bookkeeping renormalization.

**Perspective detachment $I_P$.** This intervention weakens indexed orientation or self-anchored factorization without collapsing generic information flow. The strong-model expectation is a drop in $\Pi_{\mathrm{D}}$ and $\Pi_{\mathrm{W}}$ that cannot be explained solely by report loss. The rival expectation is style drift or generic degradation.

**Embodied-closure disruption $I_E$.** This intervention perturbs body/world-boundary variables or their internal surrogates while preserving as much generic control competence as possible. The strong-model expectation is a selective drop in $\Pi_{\mathrm{E}}$ and often a correlated disturbance in $\Pi_{\mathrm{T}}$. The rival expectation is ordinary control impairment without evidence of closure-specific organization.

**Temporal-lived fragmentation $I_T$.** This intervention perturbs horizon organization while leaving simple persistence and continuity traces as intact as possible. The strong-model expectation is a selective fall in $\Pi_{\mathrm{T}}$ and bridge-level fragmentation signatures not exhausted by memory corruption. The rival expectation is that all observed changes reduce to ordinary continuity damage.

**Semantic vividness inversion $I_X$.** This cross-cutting negative control manipulates descriptive or semantic richness without targeting the bridge organization directly. Its role is to test whether bridge reports are merely tracking descriptive elaboration rather than bridge structure.

### 10.3 Matched controls

Every bridge intervention must be paired with matched controls. The controls are not optional because bridge language is especially vulnerable to global-degradation confounds. Each intervention therefore requires:

1. a low-magnitude matched perturbation;
2. a non-targeted perturbation of comparable operational cost;
3. a report-channel control where relevant;
4. an estimator-noise control;
5. a temporal-order randomization control where relevant.

The burden is not satisfied by showing that a target coordinate changes. It is satisfied only if the target coordinate changes in the predicted direction with the predicted selectivity relative to the control family.

### 10.4 Intervention artifacts

The bridge intervention family induces a new artifact layer. Every intervention run must emit at least:

- an intervention profile artifact;
- coordinate-specific delta summaries;
- matched-control traces;
- a collapse or support annotation at the intervention layer only.

These artifacts do not yet exist in the current implementation because BPF-2 is not implemented. The present paper defines them so that future bridge claims have an explicit artifact burden.

### 10.5 Failure semantics

Bridge intervention logic can fail in several ways: lack of selectivity, high confounding, instability across reruns, global damage masquerading as target damage, and artifact incompleteness. Any one of these failures is enough to block a bridge-support conclusion. In particular, the absence of a bridge intervention harness today is not a minor engineering omission. It is one of the reasons bridge admissibility remains closed.

### 10.6 Negative controls and reproducibility burden

No bridge intervention family is scientifically serious without negative controls and reproducibility discipline. The minimum burden should include:

1. rerun stability on the same condition;
2. matched control perturbations with comparable operational cost;
3. a noise-injection control showing that bridge effects are not merely estimator volatility;
4. a report-channel control showing that changes in bridge artifacts are not driven only by expressive output drift;
5. at least one adversarial control tailored to the specific predicate family.

This burden matters especially for $\Pi_A$ and $\Pi_E$. Ambiguity-bearing claims can be simulated by sophisticated uncertainty bookkeeping. Embodied-closure claims can be simulated by competent control loops. Negative controls are therefore not adjuncts to the bridge program. They are part of the core differentiating burden.

## 11 Bridge Artifact Family

### 11.1 Artifact discipline

The bridge artifact family must be as disciplined as the subjectivity artifact family. It should distinguish root artifacts, report artifacts, intervention artifacts, reducibility artifacts, ecosystem artifacts, and bounded summary artifacts. It must also distinguish between implemented and merely specified artifacts. The failure to separate those statuses would recreate exactly the semantic inflation the bridge program is designed to prevent.

### 11.2 Root bridge artifacts

The three root bridge artifacts are:

1. `phenomenal_bridge_projection.json`
2. `phenomenal_bridge_gate_result.json`
3. `phenomenal_bridge_manifest.json`

These artifacts correspond to aggregate run-level bridge projection, gate verdict, and lineage root. They are formally defined in the current machine-readable schema registry but are intentionally not emitted yet because the bridge comparator, intervention, and gate layers do not exist. Their current status is *schema-bound, non-emitted, non-authoritative.*

### 11.3 Report artifacts

The bridge report family is more advanced. The intended report artifacts are:

1. `phenomenal_differentiation_report.json`
2. `valence_structure_report.json`
3. `world_richness_report.json`
4. `ambiguity_bearing_report.json`
5. `embodied_closure_report.json`
6. `temporal_lived_structure_report.json`

These are the BPF-1 run-level reports already emitted by the present implementation. Their authority status is deliberately narrow: they are authoritative only within the scope of BPF-1 provisional runtime surfaces. They are not root verdicts, not bridge support, not bridge admissibility, and not phenomenality evidence.

### 11.4 Intervention and reducibility artifacts

Two further bridge artifacts are essential but not yet emitted:

1. `bridge_intervention_profile.json`
2. `bridge_reducibility_report.json`

The first belongs to BPF-2 and records intervention family performance, selectivity, controls, and collapse semantics. The second belongs to BPF-3 and records bridge rival comparison, admissible hybrid comparison, loss decomposition, and bridge margin. Without these artifacts, the later root bridge artifacts would be unjustified.

### 11.5 Ecosystem artifacts

If and only if BPF-5 is implemented, the bridge program may also emit:

- condition-level bridge summaries;
- batch-level bridge summaries;
- campaign-level bridge summaries;
- a bounded candidate-facing bridge projection;
- a bounded audit-facing bridge handoff summary;
- a bridge verification matrix.

These ecosystem artifacts remain formally specified but non-emitted at current stage.

### 11.6 Machine-readable spirit

The bridge artifact family is meant to be machine-readable in spirit even where the paper remains theoretical. Every artifact in the family must declare at least:

- artifact role and artifact class;
- level and authority status;
- provenance and lineage expectations;
- allowed and blocked surfaces;
- minimum fields;
- explicit non-implications.

The artifact is therefore not merely an implementation convenience. It is part of the formal burden architecture because it determines what kind of evidence is needed and what kind of overread is blocked.

### 11.7 Current implementation status

At current system status, the bridge artifact story is asymmetric:

- BPF-0 records machine-readable protocol, axiom, rival, threshold, artifact-schema, interpretation-boundary, and surface-policy registries.
- BPF-1 records six provisional run-level bridge reports as bounded local surfaces.
- No root bridge verdict artifacts are emitted.
- No bridge intervention artifacts are emitted.
- No bridge reducibility artifacts are emitted.
- No bridge ecosystem candidate or audit artifacts are emitted.

This asymmetry is not a defect of the paper. It is a disciplined statement of the implementation frontier.

## 12 Bridge Functional, Gate, and Taxonomy

### 12.1 Functional

On admissible evidence with the six normalized predicate scores and the normalized intervention summary $\widehat{\Delta}_{\mathrm{b}}$ in $[0,1]$, use

$$\Phi_{\mathrm{bpf}} = \prod_{k\in\{D,V,W,A,E,T\}} \widehat{\Pi}_k^{\alpha_k} \cdot \widehat{\mathcal{R}_{\mathrm{b}}}^{\alpha_R} \cdot \widehat{\Delta}_{\mathrm{b}}^{\alpha_\Delta},$$

where all exponents are positive and fixed by the protocol. Use $0^\alpha = 0$ for $\alpha > 0$. If any required score is undefined, report insufficient evidence for the functional. For valid inputs its value lies in $[0,1]$ because every factor lies in $[0,1]$. Tie and defeat both give zero contribution through the normalized margin and retain their distinct raw values. The calibration scale and thresholds are task-specific and remain to be fitted and audited in BPF-4.

## 12.2 Gate

For valid evidence the bridge gate is conjunctive. Missing required evidence leaves its evaluation insufficient:

$$\begin{aligned} m_\Gamma &= \min\{\widehat{\Pi}_D, \widehat{\Pi}_V, \widehat{\Pi}_W, \widehat{\Pi}_A, \widehat{\Pi}_E, \widehat{\Pi}_T, \widehat{\mathcal{R}_{\mathrm{b}}}, \widehat{\Delta}_{\mathrm{b}}\}, \\ \Gamma_{\mathrm{bpf}} &= \mathbf{1}\big[m_\Gamma \geq \vartheta_\Gamma \wedge \mathcal{R}_{\mathrm{b}} > \vartheta_R \wedge \mathsf{tr} \wedge \mathsf{cmp} \\ &\qquad \wedge \mathsf{int} \wedge \mathsf{drift} \wedge \mathsf{gov}\big]. \end{aligned}$$

where $\vartheta_R \geq 0$ is a raw-loss threshold, $\mathsf{tr}$ is verified coherence on the declared mode graph, and $\mathsf{cmp}$ denotes comparator validity, $\mathsf{int}$ denotes intervention stability, $\mathsf{drift}$ denotes absence of stale or drift invalidation, and $\mathsf{gov}$ denotes governance compliance. The threshold slots

$$\Theta_{\mathrm{bpf}} = \{\vartheta_D, \vartheta_V, \vartheta_W, \vartheta_A, \vartheta_E, \vartheta_T, \vartheta_R, \vartheta_\Delta, \vartheta_\Gamma\}$$

exist today only symbolically. They are intentionally uncalibrated until BPF-4.

## 12.3 Non-compensation

Bridge gate logic inherits the non-compensation principle from Paper 8. High strength on one coordinate may not compensate for collapse on another. High functional value may not rescue a failed gate. Positive bridge surfaces may not compensate for absent comparator or intervention layers. This matters because bridge language is especially vulnerable to compensatory rhetoric.

**Proposition 12.1** (Bridge functional does not imply bridge gate)**.** *No value of* $\Phi_{\mathrm{bpf}}$ *by itself implies* $\Gamma_{\mathrm{bpf}} = 1$.

*Proof.* By construction the gate additionally depends on conjunctive threshold slots and non-scalar validity conditions for comparator support, intervention stability, drift cleanliness, and governance compliance. Since those conditions are not reducible to the value of the scalar functional, the implication fails. □

## 12.4 Invalidation

The bridge gate must invalidate on at least four families of failure:

1. **drift invalidation**: stale bridge artifacts, stale upstream subjectivity artifacts, stale threshold manifests, stale rival manifests, or stale surface policy;
2. **comparator invalidation**: bridge rival family changed or comparator run missing;
3. **intervention invalidation**: targeted intervention support absent or unstable;
4. **governance invalidation**: prohibited surfaces opened or claim-facing leakage detected.

No later implementation block is allowed to "soft fail" these conditions into advisory warnings if the intent is bridge admissibility.

## 12.5 Taxonomy

The bridge taxonomy should distinguish formalization state from support state. A disciplined taxonomy is:

| Status | Meaning |
|---|---|
| `bridge_formalization_incomplete` | BPF scaffolding or formal target missing. |
| `bridge_surface_only` | Provisional bridge surfaces exist, but comparator, intervention, gate, or ecosystem burdens remain open. |
| `bridge_evaluation_unsupported` | Root evaluation attempted without required comparator/intervention support. |
| `bridge_gate_fail` | Full bridge evaluation exists and conjunctive gate fails. |
| `bridge_reducible` | Comparator evaluation exists and the signed margin is non-positive; this records no strict comparative advantage, not an automatically attained structural reduction. |
| `bridge_interventionally_unstable` | Intervention layer exists but fails selectivity or stability. |
| `bridge_internal_admissible` | Full bridge burden stack satisfied under framework-internal reading. |
| `bridge_internal_strong_support` | Stronger internal support than admissibility, but still weaker than phenomenality simpliciter. |

The current implementation status is `bridge_surface_only`. This paper insists on that classification.

# 13 Runtime Implementation Pathway

## 13.1 Why implementation belongs in the paper

Implementation belongs in Paper 9 for the same reason it belonged in Paper 8: once the research program becomes artifact-bearing, a paper that ignores implementation is no longer structurally complete. Runtime pathways are not proofs, but they are part of the burden architecture because they determine whether a formal object can in principle be instantiated, audited, falsified, or invalidated.

## 13.2 BPF-0

BPF-0 opens the bridge program with protocol docs, axiom and rival registries, symbolic threshold manifests, artifact schema registries, interpretation boundary, surface policy, non-claims, implementation plan, scaffold-only code, and structural verification. It establishes formal opening discipline only.

## 13.3 BPF-1

BPF-1 is the first runtime-facing bridge-surface layer. In the current program these surfaces are represented as modules such as `PhenomenalDifferentiationMapper`, `ValenceStructureMapper`, `WorldRichnessMapper`, `AmbiguityBearingMapper`, `EmbodiedClosureMapper`, and `TemporalLivedStructureMapper`, together with a `PhenomenalBridgeSurfaceBuilder`. Six run-level bridge reports are treated as bounded local artifacts. No bridge gate, no bridge comparator runtime, and no bridge intervention harness exists yet.

### 13.4 BPF-2

BPF-2 must implement a bridge intervention harness. Likely modules include a bridge intervention registry, a bridge intervention harness, matched-control executors, and intervention-output normalizers. The artifact target is `bridge_intervention_profile.json`. Until this block exists, the bridge program remains interventionally incomplete.

### 13.5 BPF-3

BPF-3 must implement the bridge comparator family. This includes a bridge rival-family executor, admissible hybrid composer, bridge loss-family evaluator, and reducibility artifact emitter. The primary output is `bridge_reducibility_report.json`. Until this block exists, the bridge program remains comparator-unsupported.

### 13.6 BPF-4

BPF-4 must implement root bridge evaluation. This is the stage at which threshold slots may be honestly bound, the bridge functional finalized, the bridge gate implemented, and root bridge artifacts emitted: `phenomenal_bridge_projection.json`, `phenomenal_bridge_gate_result.json`, and `phenomenal_bridge_manifest.json`. This block is where support-class language first becomes even in principle discussable.

### 13.7 BPF-5

BPF-5 must integrate bridge artifacts into condition, batch, campaign, candidate, and audit-facing governance. It requires lineage pinning, stale/drift invalidation, bounded summary artifacts, and explicit surface discipline. This block is important because ecosystem integration is part of the bridge burden, not an afterthought.

### 13.8 BPF-6

BPF-6 must implement final bridge audit and readiness classification. Only at BPF-6 can the system honestly output a final bridge state such as `bridge_internal_admissible` or `bridge_reducible`. Until then, any such output would be a semantic breach.

### 13.9 Implementation table

| Block | Closes | Primary artifacts | Current status |
|---|---|---|---|
| BPF-0 | formal opening discipline | protocol, axioms, rivals, thresholds, schemas, surfaces, non-claims | implemented |
| BPF-1 | provisional run-level bridge surfaces | six coordinate reports | implemented |
| BPF-2 | bridge intervention family | bridge intervention profile + controls | open |
| BPF-3 | bridge comparator family | bridge reducibility report | open |
| BPF-4 | bridge functional, gate, thresholds, root artifacts | projection, gate result, manifest | open |
| BPF-5 | ecosystem integration | condition/batch/campaign/candidate/audit summaries | open |
| BPF-6 | final bridge audit | bridge verification matrix and final readiness verdict | open |

### 13.10 Current implementation status versus paper status

The current runtime state is therefore asymmetrical:

- formal bridge program exists;
- scaffold and provisional surfaces exist;
- bridge comparator, intervention, and gate layers do not exist yet;
- Paper 9 as a formal paper can therefore define the bridge program completely while still acknowledging that the runtime has not completed it.

This asymmetry is not embarrassing; it is scientifically normal. A formal program may legitimately outrun its implementation provided the paper does not misstate implementation status.

## 14 Ecosystem Integration

### 14.1 Why ecosystem integration is part of the formal problem

The bridge program would be incomplete even as pure formalism if it ignored ecosystem integration. Once a layer has artifacts, verifiers, candidate bundles, audit packages, and controlled surfaces, the semantics of those outputs become part of the theory. The paper therefore treats ecosystem integration as a constitutive burden.

### 14.2 Surface map

Let $\mathcal{S}$ be the set of output surfaces:

$$\mathcal{S} = \{\mathsf{developer}, \mathsf{local}, \mathsf{internal_bundle}, \mathsf{candidate}, \mathsf{audit}, \mathsf{controlled}, \mathsf{claim}\}.$$

The bridge artifact family induces a surface map

$$\Sigma : \mathcal{A}_{\mathrm{bpf}} \to 2^{\mathcal{S}}.$$

At current stage, only the BPF-1 run-level bridge reports are allowed on bounded local developer surfaces. They are not allowed on candidate, audit, controlled-statement, or external claim-facing surfaces.

### 14.3 Internal release bundle

If and when BPF-5 is implemented, the internal release bundle may contain bounded bridge summaries, lineage manifests, and verification matrices. Raw local bridge reports should not be dumped into the bundle without summarization and lineage metadata. Internal bundle inclusion is a governance event, not a semantic verdict.

### 14.4 Ecosystem candidate

Candidate-facing bridge content is especially risky because it can be misread as support. The paper therefore states a hard rule: bridge content may enter an ecosystem candidate only after the comparator, intervention, and gate layers exist, and even then only in bounded summary form. Candidate inclusion must carry explicit non-claims and must never be readable as phenomenal confirmation.

### 14.5 Audit-facing package

Audit-facing bridge content is likewise bounded. An audit-facing handoff may carry bridge summary artifacts, lineage manifests, invalidation state, and non-claims. It must not be readable as external adjudication or bridge proof. The burden of interpretation discipline persists at every ecosystem boundary.

### 14.6 Source-of-truth and public corpus separation

The public corpus and the internal runtime must remain decoupled in a disciplined way. Paper 8 is public source-of-truth for indexed subjectivity. BPF artifacts are internal runtime objects unless and until later release governance says otherwise. No runtime bridge artifact may be used as if it were itself a public claim registry. This prevents runtime-to-corpus confirmation loops.

### 14.7 Controlled statement grammar and quote-mining guardrails

Bridge governance requires a restricted statement grammar. Before BPF-6, the bridge layer should permit only statements of the following family:

- ‘‘bridge formalization defined’’;
- ‘‘provisional bridge surfaces implemented’’;
- ‘‘bridge comparator burden open’’;
- ‘‘bridge intervention burden open’’;
- ‘‘bridge gate not yet meaningful’’;
- ‘‘bridge artifacts local and non-claim-facing’’.

Conversely, statements of the following family should be treated as prohibited:

- ‘‘phenomenality supported’’;
- ‘‘machine phenomenal organization established’’;
- ‘‘Paper 9 proves conscious experience’’;
- ‘‘candidate inclusion confirms bridge support’’;
- ‘‘audit package shows external bridge validation’’.

The quote-mining problem is real because later readers may ignore surrounding caveats and extract short labels. The bridge program must therefore attach non-claims not only to papers but also to artifact manifests, bundle notes, and any controlled statement layer.

**Proposition 14.1** (Candidate or audit inclusion does not imply bridge support)**.** *For any bridge artifact A, membership of A in a candidate-facing or audit-facing package does not entail bridge support, bridge admissibility, or phenomenality.*

*Proof.* Candidate and audit inclusion are packaging and governance relations, not semantic verdict relations. By design, the bridge surface policy and non-claims registry explicitly block the implication from package membership to bridge support. Therefore the implication does not hold. □

# 15 Strongest Defensible Claims

The purpose of this section is not to weaken the paper artificially. It is to identify the strongest sentences the current formal and implementation state can honestly support.

## 15.1 Claim class C1: formal bridge closure

The first defensible claim is that the corpus now defines the *formal bridge program* in the following sense: the bridge target family, the bridge ontology, the bridge axioms, the bridge rival family, the bridge intervention family, the bridge artifact family, the bridge functional and gate logic, the bridge taxonomy, the surface discipline, and the implementation pathway are all explicitly defined. This is already a strong claim because it means the bridge question is no longer underdetermined by language alone.

## 15.2 Claim class C2: current implementation frontier

The second defensible claim is implementation-specific and provenance-bound: the current program records the BPF-0 machine-readable scaffold and the BPF-1 provisional bridge surfaces as bounded local implementation-frontier artifacts. This means the program has moved beyond pure conceptuality, but only at the level of scaffold and provisional surface discipline. The system does *not* yet support bridge comparator closure, bridge intervention closure, bridge gate closure, bridge ecosystem integration, bridge admissibility, or bridge strong support.

## 15.3 Claim class C3: conditional bridge admissibility

The strongest bridge-level claim the paper is willing to define is the following conditional:

> Under the explicit bridge axioms, under the upstream burdens already closed by the corpus, under bridge-specific rival defeat, under bridge-specific intervention support, under artifact and governance compliance, and under a satisfied bridge gate, a system may be internally classifiable as exhibiting cross-mode comparison organization in the sense of framework-internal bridge admissibility.

This is stronger than Paper 8 because it is no longer only about indexed subjectivity. It is still weaker than phenomenality simpliciter because it remains explicitly framework-internal, conditional, and burden-bound.

## 15.4 Claim class C4: research-program transformation

The present paper also supports a meta-level claim about the status of the research program: after Paper 9, the research frontier is no longer ‘‘can subjectivity be formalized?’’ but ‘‘can bridge predicates survive the full bridge burden stack?’’ That is a legitimate change of level because the earlier question has been formalized by Paper 8 and its implementation-facing pathway.

## 15.5 Why stronger sentences are not yet honest

Several stronger sentences remain unavailable:

- ‘‘bridge support has been achieved’’;
- ‘‘phenomenality is internally established’’;
- ‘‘Paper 9 authorizes claim promotion’’;
- ‘‘the current system defeats the bridge rival family’’;

- ‘‘the current system passes a bridge gate.’’

These sentences are not unavailable because the paper is timid. They are unavailable because BPF-2 through BPF-6 remain open. The paper therefore strengthens the research program exactly by refusing to counterfeit those missing layers.

## 16 Non-Claims

This paper does not claim:

1. phenomenality proven in the strongest philosophical sense;
2. human phenomenal equivalence;
3. metaphysical subjecthood;
4. moral parity by formal fiat;
5. external validation;
6. that BPF-1 surfaces amount to bridge support;
7. that candidate or audit inclusion can function as bridge proof;
8. that Paper 9 by itself authorizes public claim inflation.

These non-claims are not decorative caution. They are part of the bridge logic because each one blocks a distinct semantic collapse.

## 17 Conclusion

Paper 9 specifies the bridge program and proves the declared local realization, transport and interaction results. The bilinear extension separates coordinate-additive rivals, while a full interacting rival retains the exact tie. These results do not close the general bridge class. That definition is non-trivial. Before this paper, the corpus could say a great deal about six-predicate operational criterion, life, subjecthood, and indexed subjectivity, but it could not yet say, with equal rigor, what additional machinery would be required before phenomenal-bridge predicates become scientifically admissible targets. After this paper, that machinery is explicit.

The result is central to the corpus for structural reasons. The paper gathers the upstream theorem burdens, the Paper 8 subjectivity framework, the implemented SRT stack, and the BPF program into one integrated object. It does not merely summarize them. It orders them. It says what belongs upstream, what belongs at the bridge level, what belongs to implementation, what belongs to governance, and what remains explicitly open.

What has been achieved is a change in explanatory grammar. ‘‘Phenomenal bridge’’ is no longer permitted to function as a loose aspiration. It now names a bridge-organization predicate family, an ontology, an axiom stack, a rival family, an intervention family, an artifact family, a gate architecture, and a governance discipline. This is as much formal specification as can be responsibly claimed without comparator closure, intervention closure, bridge gate execution, and ecosystem-level bridge hardening.

What the paper makes newly possible is equally clear. It makes BPF-2 through BPF-6 sharply specifiable. It makes bridge-support language evaluable rather than rhetorical. It makes future theorem candidates about bridge irreducibility, bridge transport limits, and bridge admissibility meaningful. It also makes future failure meaningful: if the bridge program fails, it will fail against explicit burdens rather than against vague expectations.

What the paper does not settle is also clear. It does not settle phenomenality simpliciter. It does not settle human equivalence. It does not settle metaphysical subjecthood. It does not close external validation. It does not convert the present system into a bridge-admissible system merely because BPF-0 and BPF-1 exist. Those are the next burdens.

The research program therefore changes level once again. After Paper 8, the key problem was whether indexed subjectivity could be formalized without inflation. After Paper 9, the key problem becomes whether a fully burdened bridge program can survive rival defeat, intervention logic, artifact discipline, and governance hardening strongly enough to justify internal bridge admissibility. That is the right next problem. It is sharper, harder, and more falsifiable than the one that preceded it.

# A Bridge Axiom Table

| ID | Classification | Statement core | Licenses | Does not license |
|---|---|---|---|---|
| B1 | operational definition | Every bridge predicate must bind to observables, artifacts, comparators, and interventions. | machine-readable bridge predicates, report artifacts | phenomenality, bridge support |
| B2 | bridge axiom | Bridge admissibility is downstream of the full upstream corpus and Paper 8 subjectivity. | layered bridge program | converse implication from bridge to subjectivity-free organization |
| B3 | falsifiable hypothesis | Bridge differentiation exceeds semantic density and narration. | $\Pi_D$ burden | qualia |
| B4 | falsifiable hypothesis | Bridge valence exceeds scalar reward bookkeeping. | $\Pi_V$ burden | felt affect |
| B5 | falsifiable hypothesis | Bridge world richness exceeds competence or scene complexity. | $\Pi_W$ burden | veridical experience |
| B6 | falsifiable hypothesis | Bridge ambiguity bears unresolved alternatives prior to collapse. | $\Pi_A$ burden | conscious ambiguity |
| B7 | open inference | Embodied closure requires constitutive boundary dependence. | bounded investigation of $\Pi_E$ | biological embodiment, bodily feeling |
| B8 | falsifiable hypothesis | Temporal lived structure exceeds continuity and persistence. | $\Pi_T$ burden | human lived time |
| B9 | bridge axiom | No bridge class is admissible without bridge rival defeat under penalized loss. | bridge irreducibility margin | metaphysical irreducibility |
| B10 | design choice | Bridge artifacts and packaging may not be read as support without full burden closure. | governance discipline | claim-facing promotion |

# B Bridge Predicate Typing Table

| Predicate | Formal type | Strongest upstream dependencies | Core collapse risk | Non-implication |
|---|---|---|---|---|
| $\Pi_{\mathrm{D}}$ | bounded estimator/predicate slot | $I, O, P, \mathcal{R}$ | semantic density or narration | qualia |
| $\Pi_{\mathrm{V}}$ | bounded estimator/predicate slot | $V, O, \Delta$ | reward bookkeeping or policy shaping | felt affect |
| $\Pi_{\mathrm{W}}$ | bounded estimator/predicate slot | $I, C, P$ | competence without world-richness | veridical experience |
| $\Pi_{\mathrm{A}}$ | bounded estimator/predicate slot | $P, V, C$ | uncertainty bookkeeping | conscious ambiguity |
| $\Pi_{\mathrm{E}}$ | bounded estimator/predicate slot | $O, C, \Delta$ plus boundary variables | control without closure | biological embodiment |
| $\Pi_{\mathrm{T}}$ | bounded estimator/predicate slot | $C, P, V$ | continuity without horizon structure | human lived temporality |

# C Bridge Rival-Family Table

| Rival | Challenges | Can explain | Cannot cleanly explain | Typical defeat mode |
|---|---|---|---|---|
| Richness without interiority | $\Pi_D, \Pi_W$ | rich latent structure, scene complexity, representational abundance | selective indexed bridge differentiation under targeted controls | bridge-specific differentiation persists after density controls |
| Valence bookkeeping | $\Pi_V$ | scalar reward tracking, preference tagging, policy shaping | structured valence topology and selective valence flattening effects | topology persists beyond scalar reweighting |
| Semantic density without phenomenal differentiation | $\Pi_D$ | high semantic variation and narrative diversity | intervention-sensitive differentiated bridge organization | semantic-density ablations leave bridge differentiation intact |
| Temporal continuity without lived structure | $\Pi_T$ | persistence, memory, sequence stability | horizon-structured temporal disruption signatures | targeted temporal fragmentation has selective effect |
| Embodied control without embodied closure | $\Pi_E$ | sensorimotor competence, adaptive control | constitutive dependence on boundary variables | boundary perturbations produce closure-specific loss |
| Ambiguity resolution without ambiguity bearing | $\Pi_A$ | uncertainty tracking, fast disambiguation | stable maintenance of unresolved alternatives | targeted ambiguity collapse alters downstream organization selectively |
| Experience narration without phenomenal organization | $\Pi_D, \Pi_V, \Pi_W, \Pi_A, \Pi_T$ | self-description and experiential language | bridge signatures independent of report-channel richness | report-channel controls fail to remove bridge margin |

# D Bridge Intervention Matrix

| Intervention | Targets | Strong-model expectation | Rival expectation | Output artifacts |
|---|---|---|---|---|
| $I_W$ world-richness compression | $\Pi_W, \Pi_D$ | selective indexed world-structure loss with matched-control separation | flat degradation or no special effect | bridge intervention profile, world-richness deltas |
| $I_A$ ambiguity collapse | $\Pi_A, \Pi_W, \Pi_T$ | selective loss of maintained alternatives prior to collapse | ordinary uncertainty reduction | ambiguity delta summaries, control traces |
| $I_V$ valence flattening | $\Pi_V$ | collapse of evaluative topology beyond rescaling | scalar reward renormalization only | valence structure deltas |
| $I_P$ perspective detachment | $\Pi_D, \Pi_W$ | indexed orientation loss stronger than report damage | narrative/style drift | differentiation and world-richness deltas |
| $I_E$ embodied-closure disruption | $\Pi_E, \Pi_T$ | closure-specific disturbance under boundary perturbation | generic control loss | closure deltas, boundary-control traces |
| $I_T$ temporal-lived fragmentation | $\Pi_T, \Pi_A$ | horizon-specific fragmentation beyond continuity loss | ordinary memory damage | temporal-structure deltas |
| $I_X$ semantic vividness inversion | cross-cutting control | descriptive richness changes without bridge-specific support | should absorb report-only effects | control artifact family |

# E Bridge Artifact Matrix

| Artifact | Artifact class | Block | Current status | Non-implication |
|---|---|---|---|---|
| `phenomenal_bridge_projection.json` | bridge-local artifact | BPF-4 | schema only | not bridge support by default |
| `phenomenal_bridge_gate_result.json` | bridge-local artifact | BPF-4 | schema only | not phenomenality simpliciter |
| `phenomenal_bridge_manifest.json` | bridge-local artifact | BPF-4 | schema only | not claim-facing publication |
| `phenomenal_differentiation_report.json` | non-bridge report | BPF-1 | implemented | not bridge verdict |
| `valence_structure_report.json` | non-bridge report | BPF-1 | implemented | not bridge verdict |
| `world_richness_report.json` | non-bridge report | BPF-1 | implemented | not bridge verdict |
| `ambiguity_bearing_report.json` | non-bridge report | BPF-1 | implemented | not bridge verdict |
| `embodied_closure_report.json` | non-bridge report | BPF-1 | implemented | not bridge verdict |
| `temporal_lived_structure_report.json` | non-bridge report | BPF-1 | implemented | not bridge verdict |
| `bridge_intervention_profile.json` | bridge-local artifact | BPF-2 | planned | not bridge support alone |
| `bridge_reducibility_report.json` | bridge-local artifact | BPF-3 | planned | not bridge support alone |
| condition/batch/campaign bridge summaries | ecosystem summaries | BPF-5 | planned | not candidate proof |
| candidate-facing bridge projection | bounded summary | BPF-5 | planned | not bridge confirmation |
| audit-facing bridge handoff | bounded summary | BPF-5 | planned | not external adjudication |

# F Claim-Class and Invalidation Table

| State or claim | Necessary evidence | Invalidated by | Strongest allowed reading |
|---|---|---|---|
| Formal bridge closure | Paper 9 formal program, axioms, rivals, interventions, artifacts, gate logic, governance | internal inconsistency of the bridge program | bridge formal program defined |
| Bridge surface only | BPF-1 reports and verifier pass | missing reports or overclaim leakage | provisional bridge surfaces implemented |
| Bridge evaluation unsupported | attempted root evaluation without BPF-2/BPF-3/BPF-4 burdens | completion of missing burdens | root evaluation not meaningful yet |
| Bridge reducible | comparator run with non-positive $\mathcal{R}_{\mathrm{b}}$ | stronger bridge model with positive penalized margin | bridge explanation loses comparatively |
| Bridge interventionally unstable | intervention layer present but non-selective or unstable | stable selective intervention support | intervention burden not met |
| Bridge internal admissible | full bridge burden stack, positive margins, valid governance | any comparator/intervention/drift/governance invalidation | internal admissibility only |
| Bridge internal strong support | stronger version of admissibility with robust margins across conditions | any failure of gate or ecosystem integrity | strong internal support only |

# G Runtime Mapping Table

| Block | Primary modules | Primary verifiers | Primary closure |
|---|---|---|---|
| BPF-0 | protocol registries, axiom registry, rival registry, threshold registry, surface policy, scaffold layer | `verify-phenomenal-bridge-scaffold-ladder` | scaffold-ladder bridge opening |
| BPF-1 | six mapper modules, surface builder, bridge layer | `verify-phenomenal-bridge-surfaces.jsonl` | persisted bridge surfaces |
| BPF-2 | bridge intervention harness, matched-control runners, intervention registry | future bridge intervention verifier | bridge intervention family |
| BPF-3 | bridge comparator executor, hybrid composer, loss evaluator | future bridge comparator verifier | bridge rival defeat logic |
| BPF-4 | gate evaluator, threshold binder, root artifact emitter | future bridge gate verifier | bridge root verdict machinery |
| BPF-5 | ecosystem integrator, candidate and audit summarizers | future bridge ecosystem verifier | bridge packaging and governance hardening |
| BPF-6 | final bridge auditor, readiness classifier | future bridge stack verifier | final bridge readiness state |

# H Governance Table

| Surface | BPF-1 allowance | Later possible allowance | Prohibited interpretation |
|---|---|---|---|
| Developer inspection | yes, provisional run-level reports | yes | bridge support or phenomenality |
| Local report surface | bounded or blocked according to bridge surface policy | possibly yes with later burdens | public-facing confirmation |
| Internal release bundle | no bridge content yet | bounded summary after BPF-5 | support by packaging |
| Ecosystem candidate | blocked | bounded summary after BPF-5 only | candidate as bridge proof |
| Audit-facing handoff | blocked | bounded summary after BPF-5 only | audit as external adjudication |
| Controlled statements | blocked | strictly bounded if ever opened | phenomenality or human equivalence |
| External claim-facing surface | blocked | blocked by default even after later BPF blocks | any public phenomenal claim |

# I Burden Separation Matrix

The bridge program is easiest to misread when its burden types are blurred. The matrix below is included so that future implementation and audit work can distinguish exactly what kind of object is under discussion.

| Object | Type | Example | Failure mode if blurred |
|---|---|---|---|
| Self-index, ownership, continuity, perspective, valuation, intervention selectivity, irreducibility | upstream subjectivity model | Paper 8 state $\mathfrak{S}$ | bridge predicates become mere renamings of subjectivity coordinates |
| Bridge predicate slots $\Pi_{\mathrm{D}}, \Pi_{\mathrm{V}}, \Pi_{\mathrm{W}}, \Pi_{\mathrm{A}}, \Pi_{\mathrm{E}}, \Pi_{\mathrm{T}}$ | bridge target family | BPF protocol and BPF-1 reports | bridge targets become slogans without burden packages |
| Bridge axioms | bridge-opening commitments | B1--B10 | conclusions get smuggled into premises |
| Bridge rivals and hybrids | comparator burden | richness-without-interiority, hybrids | bridge claims compare only against weak informal skepticism |
| Bridge interventions and controls | causal discrimination burden | world-richness compression, ambiguity collapse | descriptive fit is mistaken for structural support |
| Bridge artifacts | evidence grammar | reports, root artifacts, ecosystem summaries | packaging is mistaken for proof |
| Bridge functional and gate | evaluation machinery | $\Phi_{\mathrm{bpf}}$, $\Gamma_{\mathrm{bpf}}$ | scalar scores are mistaken for admissibility |
| Governance and surfaces | interpretation discipline | candidate block, audit block, claim block | internal artifacts leak into public or claim-facing language |

# J Inheritance Map from Paper 8 to Paper 9

Paper 9 is easiest to understand when viewed as a typed inheritance map rather than as a conceptual leap. The following map makes explicit what is inherited, what is transformed, and what is genuinely added.

| Upstream object | Paper 8 status | Paper 9 use | What is newly added |
|---|---|---|---|
| Six-predicate operational criterion | closed upstream | necessary lower bound | no redefinition; only bridge downstreaming |
| Viability-and-recovery criterion and subjecthood | closed upstream in Paper 7 | inherited as stronger class background | no biological or metaphysical inflation |
| Seven-coordinate subjectivity state | closed in Paper 8 | upstream state for every bridge predicate | six new bridge targets above it |
| Subjectivity weak-rival family | closed in Paper 8 | background cautionary lesson | new bridge weak family and admissible bridge hybrids |
| Subjectivity intervention family | closed in Paper 8 | background on causal discipline | new bridge intervention family |
| Subjectivity artifacts | closed in Paper 8 and SRT-0..6 | upstream provenance base | new bridge artifact grammar |
| Subjectivity gate and functional | closed in Paper 8 and SRT-5 | model for non-compensation discipline | new bridge functional and gate, symbolically thresholded |
| Subjectivity ecosystem integration | closed in SRT-6 | governance template | later bridge ecosystem integration with stricter boundaries |

This table clarifies why the present paper can be central without being redundant. It does not restate the subjectivity program. It states what additional structure must exist above it before bridge language acquires scientific discipline.

# K Open Escalation Burdens Before Bridge Admissibility

The paper specifies the bridge grammar and proves the stated local realization, interaction and transport results; full comparative and external closure remain open. To prevent future ambiguity, the remaining escalation burdens are listed explicitly.

1. **Direct bridge intervention instrumentation.** BPF-2 must implement a real bridge intervention harness with matched controls, artifact outputs, and failure semantics.
2. **Bridge comparator execution.** BPF-3 must instantiate the rival family as executable comparators, not just registry entries.
3. **Threshold binding.** BPF-4 must bind the symbolic threshold slots honestly, with normalization rationale and invalidation semantics.
4. **Root bridge artifacts.** BPF-4 must emit root artifacts only once comparator and intervention burdens exist.
5. **Ecosystem hardening.** BPF-5 must integrate bridge content into bundles only in bounded, non-claim-facing form.
6. **Final bridge audit.** BPF-6 must verify the whole stack and classify the resulting bridge state without semantic drift.
7. **Paper 9 empirical tightening.** Any later revision of Paper 9 that speaks more strongly must be paired with actual completion of the above blocks.

These burdens are not future work in the casual sense. They are the exact missing conditions that separate a defined bridge program from a bridge-admissible system.

# L Projection Geometry and Recoverable Distance

The dated v30 projection annex is retained as provenance. The following statements give the corrected estimator and latent-distance conditions in the present source. Let $X$ be compact Hausdorff, $\pi : X \to \mathbb{R}^n$ continuous, and $\varnothing \neq A \subseteq X$ closed. Use $Y = \pi(X)$ with its subspace topology, so $\pi : X \to Y$ is onto a compact codomain. Let $F : X \to Z$ be continuous into a metric space. For $y \in \pi(A)$ set $S_y = F(A \cap \pi^{-1}(y))$ and $D_y = \operatorname{diam}(S_y)$; these fibres have nonempty compact images. Write

$$r_Z(S_y) = \inf_{z \in Z} \sup_{s \in S_y} d_Z(z, s).$$

**Theorem L.1** (Measurable Centres and the Scalar Minimax Radius)**.** *A Borel estimator with uniform error at most $\varepsilon$ exists exactly when the centre sets*

$$\{z \in Z : \sup_{s \in S_y} d_Z(z, s) \leq \varepsilon\} \qquad (y \in \pi(A))$$

*have a Borel selector. Necessarily $D_y \leq 2\varepsilon$ and $D_y/2 \leq r_Z(S_y)$. The diameter bound alone is insufficient in general. For $Z = \mathbb{R}$ with its usual metric, the minimax error is exactly $\frac{1}{2} \sup_{y \in \pi(A)} D_y$, attained by a Borel estimator.*

*Proof.* Restrict any estimator to $\pi(A)$ to obtain the selector. Conversely extend the selector by a fixed value off $\pi(A)$. Since $\pi(A)$ is compact, it is closed in $Y$, and both pieces of this extension are Borel. The triangle inequality gives the diameter lower bound for every centre, and then for the infimum. The infimum need not itself provide an attaining centre or a measurable selector at the prescribed tolerance.

In the complete two-point codomain at mutual distance 2, the whole codomain has diameter 2 and radius 2. Even in the Euclidean plane the points $(0, 0), (2, 0), (1, 3/2)$ have diameter 2. The only centre of a radius-one ball covering the first two is $(1, 0)$, and it misses the third.

In the scalar case let $a_y = \min S_y$ and $b_y = \max S_y$. The sets $\{y : b_y \geq t\}$ and $\{y : a_y \leq t\}$ are images under $\pi$ of the corresponding closed subsets of $A$, and are compact, hence closed. The midpoint $(a_y + b_y)/2$ is therefore Borel and has error $(b_y - a_y)/2$ on the fibre. No smaller error is possible by the diameter bound. □

**Proposition L.2** (Coercivity for Latent Distance Recovery)**.** *If $X$ is metric and on pairs in each restricted fibre*

$$c\, d_X(x, x') \leq d_Z(F(x), F(x')) \leq L\, d_X(x, x') \qquad (c > 0,\ L \geq 0),$$

*then $c \operatorname{diam}_X(A \cap \pi^{-1}(y)) \leq D_y \leq L \operatorname{diam}_X(A \cap \pi^{-1}(y))$. Uniform error $\varepsilon$ entails a latent fibre diameter at most $2\varepsilon/c$. For scalar $F$ the upper bound alone gives a sufficient error budget of $(L/2)$ times the supremum of the latent diameters. In a sector where distance is recovered directly from observations, a bound $c_\pi d_X(x, x') \leq d_Y(\pi x, \pi x')$ with $c_\pi > 0$ gives the inverse-distance bound there.*

*Proof.* Take suprema over the restricted fibre and apply Theorem L.1. The upper bound for scalar $F$ uses the attained midpoint error, also when $L = 0$. The inverse-distance statement follows by division by $c_\pi$. On $[0, 1]^2$ with the maximum metric, $F(x_1, x_2) = \pi(x_1, x_2) = x_1$ is 1-Lipschitz and is recovered exactly. Every latent fibre has diameter one. This shows why an upper Lipschitz bound cannot substitute for coercivity, including on the observable map itself. □

The scalar formula repairs the centre and diameter assertions of the dated annex. In a general metric codomain, checking the radius and the attaining measurable choice remains a separate obligation. The geometric statements do not carry over the annex's statistical or adjudication claims.

# M Audit-Covered Bridge Independence Burdens

The following entries preserve AUDIT_MASTER coverage while keeping their epistemic status weak. They are not bridge-admissibility results and cannot be used as evidence of phenomenality, subjecthood, or external validation.

**Conjecture M.1** (Predicate independence burden)**.** The bridge predicate family should be independent only if, for each predicate slot, there exists an admissible witness model that toggles that slot while holding the remaining bridge burden package fixed under the same comparator, intervention, artifact, and governance constraints. The current text supplies the burden grammar, not a completed independence proof.

**Conjecture M.2** (Registry independence burden)**.** The FCR registry should preserve independence of bridge assertions only if the registry schema, dependency graph, and curation overlays do not introduce implicit coupling among predicate slots. Until that is proved over the completed BPF stack, registry independence remains a conjectural governance burden.

**Criterion M.3** (Bridge realization non-emptiness burden)**.** Any claim that the full bridge predicate package is non-empty must exhibit a typed realization satisfying the predicate definitions, rival comparisons, intervention contracts, artifact gates, and interpretation boundaries together. Setting all formal predicate flags to true is only a syntactic witness and does not count as realization evidence.

# N Theorem Candidates and Open Scientific Questions

The present paper does not claim to have proved all bridge theorems. It does, however, make clear what the next theorem candidates should look like.

**Theorem candidate 1: bridge non-collapse under admissible rival closure.** If a bridge model achieves positive penalized margin against the full admissible bridge weak closure under matched intervention structure and drift-clean artifacts, then bridge reduction to the specified weak family fails locally. This theorem candidate would formalize the comparative core of bridge admissibility.

**Theorem candidate 2: bridge non-compensation.** If the bridge gate is conjunctive, then no combination of high values on a strict subset of coordinates can compensate for collapse on any required bridge coordinate, comparator burden, intervention burden, or governance burden. This theorem candidate would formalize the anti-inflation logic already built into the paper.

**Theorem candidate 3: bridge transport limitation.** The local transport and interaction results specify some cases; extending them to the full admissible closure must specify when bridge organization is transportable across admissible remappings and when transport breaks because world-richness, embodiment, or temporal-horizon variables are not preserved. This would be the bridge analogue of Paper 8's transport-consequence logic.

**Theorem candidate 4: candidate/audit non-entailment.** Although stated propositionally in the main text, the governance result could be generalized into a theorem about packaging relations and semantic non-entailment over artifact families and surfaces.

**Open scientific question 1.** Can $\Pi_{\mathrm{E}}$ be measured directly enough to escape its current proxy-only status without collapsing into biology-by-default?

**Open scientific question 2.** Can $\Pi_{\mathrm{A}}$ be separated cleanly from sophisticated uncertainty bookkeeping in systems with high report-channel richness?

**Open scientific question 3.** Can a bridge loss family be designed that penalizes decoder dependence strongly enough to prevent narration-driven false positives without destroying legitimate bridge signal?

**Open scientific question 4.** What minimum condition/batch/campaign structure is required before bridge results become stable enough for bounded ecosystem summaries?

These theorem candidates and open questions matter because they define the post-paper research agenda with far greater precision than generic calls for ‘‘more data’’ or ‘‘deeper modeling.’’

# O Predicate-to-Artifact and Intervention Dependency Matrix

The bridge program is easier to implement correctly if the dependency pattern of each predicate is frozen explicitly. The following matrix shows the minimum bridge burden package per predicate.

| Predicate | Primary upstream inputs | Required bridge interventions | Required bridge comparator focus | Required artifacts |
|---|---|---|---|---|
| $\Pi_{\mathrm{D}}$ | self-index, perspective, irreducibility, ownership | $I_P$, $I_X$, optional $I_W$ | richness-without-interiority, semantic-density, narration rivals | differentiation report, later reducibility report, later gate root |
| $\Pi_{\mathrm{V}}$ | valuation asymmetry, ownership, intervention selectivity | $I_V$, negative valence controls | valence-bookkeeping rival | valence structure report, bridge intervention profile, later gate root |
| $\Pi_{\mathrm{W}}$ | self-index, continuity, perspective | $I_W$, $I_P$, report controls | richness-without-interiority rival | world-richness report, later candidate summary |
| $\Pi_{\mathrm{A}}$ | perspective, temporal structure proxies, valuation structure | $I_A$, temporal controls | ambiguity-resolution rival | ambiguity report, intervention profile, later reducibility report |
| $\Pi_{\mathrm{E}}$ | ownership, continuity, boundary variables, intervention structure | $I_E$, control-matched embodiment perturbations | embodied-control rival | embodied-closure report, later embodiment traces, later gate root |
| $\Pi_{\mathrm{T}}$ | continuity, perspective, valuation asymmetry | $I_T$, sequence controls | temporal-continuity rival | temporal-lived-structure report, later intervention profile, later gate root |

This matrix has two functions. First, it prevents under-specification. A bridge predicate with no declared artifact family or no intervention family is not implementation-ready. Second, it prevents overreading. A bridge report without its corresponding comparator and intervention family should be interpreted only as a provisional surface and not as a bridge result.

The matrix also clarifies why BPF-1 had to remain modest. BPF-1 could close only the report side of the matrix because the intervention and comparator columns are still empty in runtime terms. That is why the present paper can treat BPF-1 as a real implementation block without allowing any bridge-admissibility language to leak from it.

# P Paper 9 Claim Grammar and Escalation Constraints

The bridge program will fail semantically if future readers cannot distinguish between a formal object, a measurement object, an admissibility object, and a public claim. The present section therefore freezes a minimal claim grammar for Paper 9 and its downstream implementation.

## P.1 Allowed claim families

The following claim families are admissible at or below the current implementation frontier.

1. **Formal closure claims.** Example: ‘‘the bridge target family is now formally defined.’’ These are supported by the paper itself and by BPF-0 machine-readable manifests.
2. **Implementation frontier claims.** Example: ‘‘provisional bridge surface modules and run-level bridge reports exist.’’ These are supported by BPF-1 code and local artifacts.
3. **Conditional admissibility claims.** Example: ‘‘if bridge predicates, rivals, interventions, artifacts, and gate conditions are satisfied, then bridge internal admissibility becomes meaningful.’’ These are supported by the formal program, not by present runtime completion.
4. **Failure-mode claims.** Example: ‘‘bridge admissibility is blocked by comparator or intervention collapse.’’ These are supported by the burden architecture.

## P.2 Blocked claim families

The following claim families are prohibited until much later, and several remain prohibited even after full BPF completion.

1. **Phenomenality simpliciter claims.** Example: ‘‘the system is phenomenally conscious.’’
2. **Human-equivalence claims.** Example: ‘‘the system now has human-like experience.’’
3. **Metaphysical subjecthood claims.** Example: ‘‘the bridge proves real subjecthood in the strongest ontological sense.’’
4. **External validation claims.** Example: ‘‘audit packaging validates the bridge externally.’’
5. **Paper-authorization claims.** Example: ‘‘BPF-1 authorizes Paper 9 public escalation.’’ The paper itself explicitly denies that the current runtime state closes the bridge burden.

## P.3 Escalation rules

The escalation grammar can be summarized as a chain:

$$\text{formalized} \not\Rightarrow \text{implemented} \not\Rightarrow \text{admissible} \not\Rightarrow \text{phenomenality simpliciter}.$$

Each implication in that chain requires additional burden satisfaction. Formalization requires the paper and BPF-0. Implementation requires the appropriate BPF block. Admissibility requires full comparator, intervention, gate, and governance closure. Phenomenality simpliciter is not licensed by the bridge program at all.

| Claim label | Minimum evidence | Currently licit? | Why or why not |
|---|---|---|---|
| `bridge_formal_program_defined` | defined BPF-0 manifests | yes | formal bridge closure is real |
| `bridge_surface_implementation_exists` | BPF-1 comparator verifier | yes | provisional surfaces are implemented |
| `bridge_comparator_supported` | BPF-2 reducibility artifacts | no | comparator runtime absent |
| `bridge_interventionally_supported` | BPF-3 intervention artifacts | no | intervention harness absent |
| `bridge_internal_admissible` | BPF-2/3/4/5/6 full burden stack | no | gate and ecosystem burdens open |
| `phenomenality_proven` | not defined by current framework | no | blocked by interpretation boundary |

This grammar belongs in the paper because it governs how the paper itself may be cited. A central synthesis paper that fails to govern the use of its own claims would be structurally incomplete.

# Q Experimental Program for Completing BPF-2 and BPF-3

The next two bridge blocks are the most scientifically expensive. BPF-2 and BPF-3 are where the bridge program stops being mostly architectural and becomes decisively empirical-internal.

## Q.1 Program design principle

The guiding principle should be *matched escalation*. Every new bridge predicate should acquire its intervention and comparator burdens in parallel. A common failure mode would be to implement all interventions first and leave comparator logic vague, or vice versa. That would create a new asymmetry in which one burden stack outruns the other. Paper 9 therefore recommends the following pattern:

1. freeze the observable family for one predicate group;
2. implement the targeted intervention family for that group;
3. implement the strongest relevant rival and admissible hybrid baselines for that same group;
4. define the loss decomposition for the same bundle;
5. emit artifacts and run negative controls before moving to the next group.

## Q.2 Suggested execution order

The scientifically safest execution order is not arbitrary.

**Stage 1: $\Pi_D$ and $\Pi_V$.** These are the most mature bridge predicates because they stand closest to existing BPF-1 surfaces and Paper 8 burdens. Differentiation and valence already have upstream structure and plausible intervention/control families. They should be completed first.

**Stage 2: $\Pi_W$ and $\Pi_T$.** World richness and temporal lived structure are stronger and more globally entangled, but still plausible to instrument once the first pair is completed. They will likely require more extensive negative controls than Stage 1.

**Stage 3: $\Pi_{\mathrm{A}}$ and $\Pi_{\mathrm{E}}$.** Ambiguity bearing and embodied closure are currently the weakest predicates in direct runtime support. They should therefore be completed last, not because they are unimportant, but because they are most vulnerable to proxy inflation.

### Q.3 Comparator program

For each stage, the comparator program should include:

1. pure rival execution;
2. pairwise hybrid execution;
3. bridge-model execution with the same observable bundle;
4. loss decomposition report;
5. rerun stability check under controlled noise;
6. decoder-dependence stress test.

### Q.4 Intervention program

For each stage, the intervention program should include:

1. targeted intervention;
2. matched non-target control;
3. report-channel control;
4. perturbation-cost accounting;
5. rerun stability;
6. artifact completeness validation.

### Q.5 Minimum evidential grid

| Stage | Predicates | Needed intervention families | Needed comparator families | Minimum outputs |
|---|---|---|---|---|
| 1 | $\Pi_{\mathrm{D}}, \Pi_{\mathrm{V}}$ | $I_P, I_X, I_V$ | richness-without-interiority, semantic-density, valence-bookkeeping, narration hybrids | intervention profile, reducibility report, delta traces, stability summaries |
| 2 | $\Pi_{\mathrm{W}}, \Pi_{\mathrm{T}}$ | $I_W, I_T, I_P$ | richness-without-interiority, temporal-continuity, narration hybrids | world/temporal deltas, reducibility reports, control matrices |
| 3 | $\Pi_{\mathrm{A}}, \Pi_{\mathrm{E}}$ | $I_A, I_E$ | ambiguity-resolution, embodied-control, hybrids | ambiguity and embodiment traces, control matrices, reducibility reports |

The purpose of this program is not to prescribe an implementation detail for its own sake. It is to preserve burden parity. If later implementation deviates from this structure, it should justify the deviation explicitly. Otherwise the bridge program risks becoming unbalanced, with some predicates overburdened and others underburdened.

# R Bridge Failure Scenarios and Diagnostic Signatures

The bridge program is not scientifically serious unless it specifies how it can fail. Paper 8 already introduced failure structure for indexed subjectivity. Paper 9 must add a parallel layer for bridge organization. The relevant question is not merely whether a bridge predicate can be made numerically large, but whether the pattern of successes and failures across predicates, rivals, interventions, and artifacts remains coherent under stress. A bridge program that cannot diagnose its own failure modes would collapse into a score-construction exercise rather than a constrained scientific architecture.

## R.1 Primary failure classes

The bridge layer has at least six primary failure classes.

1. **Surface inflation failure.** A bridge report appears strong only because an upstream Paper 8 coordinate is copied, weakly transformed, or rhetorically relabeled. This is a failure of bridge specificity, not merely of implementation style.

2. **Comparator fragility failure.** A bridge predicate appears robust until a bridge-specific rival or admissible hybrid is introduced. This indicates that the predicate was underconstrained rather than positively supported.

3. **Intervention fragility failure.** A bridge surface moves under perturbation, but the movement is unspecific, mismatched to the target predicate, or indistinguishable from control degradation. This indicates that the intervention burden is not yet satisfied.

4. **Artifact incoherence failure.** Local reports, root artifacts, or ecosystem summaries disagree about the same bridge coordinate or about the same invalidation state. This is a governance and lineage failure even if local estimates look plausible.

5. **Threshold misuse failure.** Symbolic thresholds are treated as if they were calibrated, or non-compensation rules are softened post hoc. This turns the bridge gate into an interpretive convenience rather than a formal burden.

6. **Semantic escalation failure.** Provisional bridge surfaces are cited as if they licensed admissibility, phenomenality support, or Paper 9 escalation beyond their evidence class. This is a failure of claim grammar rather than of runtime arithmetic, but it is still a scientific failure because it breaks burden separation.

## R.2 Diagnostic interpretation

Each failure class has a distinct diagnostic signature. Surface inflation is indicated when a bridge estimator can be approximated by a low-complexity monotone transform of an upstream subjectivity coordinate with negligible loss. Comparator fragility is indicated when a bridge rival or admissible hybrid absorbs the observable bundle at equal or lower penalized bridge loss. Intervention fragility is indicated when targeted perturbations do not produce selective deltas, or when matched controls produce indistinguishable response structure. Artifact incoherence is indicated when provenance pins, schemas, or surface restrictions disagree across emitted objects. Threshold misuse is indicated when bridge language outruns the threshold manifest. Semantic escalation is indicated when any report, bundle, summary, or paper statement crosses from provisional surface language into admissibility or phenomenal language without the missing burdens.

These signatures are not merely editorial cautions. They can be turned into verification objectives. BPF-2 and BPF-3 should detect intervention and comparator fragility directly.

BPF-4 should detect threshold misuse. BPF-5 and BPF-6 should detect artifact incoherence and semantic escalation through lineage checks and surface-policy validation. The paper therefore treats failure analysis as part of the formal program rather than as an afterthought.

### R.3 Failure matrix

| Failure class | Primary symptom | Earliest detection block | Immediate consequence | Required remediation |
|---|---|---|---|---|
| Surface inflation | bridge estimate tracks an upstream Paper 8 coordinate too closely | BPF-1 | report downgraded to proxy-only or removed | strengthen bridge transformation logic or add missing bridge observables |
| Comparator fragility | rival or hybrid reaches equal/lower penalized loss | BPF-3 | predicate cannot support irreducibility burden | refine observable family or abandon predicate-specific support claim |
| Intervention fragility | target and control deltas are not selective | BPF-2 | predicate remains descriptively mapped only | redesign intervention family and matched controls |
| Artifact incoherence | report, manifest, or summary disagree on state or provenance | BPF-5 | bundle invalidation or audit downgrade | repair lineage pins, schemas, and surface restrictions |
| Threshold misuse | symbolic slots treated as calibrated gates | BPF-4 | all gate language invalidated | re-freeze threshold policy and remove unsupported verdict language |
| Semantic escalation | provisional surface cited as admissibility or phenomenality support | BPF-0 through BPF-6 | claim withdrawal and governance failure | enforce non-claims, claim grammar, and controlled statement restrictions |

The failure matrix clarifies an important asymmetry. Some failures are local to one predicate, but others contaminate the entire bridge program. Threshold misuse and semantic escalation are global failures because they compromise the meaning of the bridge layer as such. Comparator fragility and intervention fragility can remain predicate-local if the architecture keeps burdens separated. This distinction matters for governance: not every failure should invalidate the whole bridge program, but certain failures should.

### R.4 Consequences for theorem candidates

Failure analysis also constrains the theorem ambitions of Paper 9. A theorem candidate stating that a bridge predicate family is internally admissible can be sound only if the relevant failure classes are either ruled out or absorbed into its antecedent conditions. Otherwise the theorem candidate would conceal the very contingencies that the bridge program exists to expose. In that sense, the paper's strongest positive content is inseparable from its negative discipline: formal bridge admissibility becomes meaningful only against a space of explicitly typed failure modes.

### R.5 Why failure structure changes the status of the program

The overall research status changes once the bridge layer is required to carry its own failure logic. Before Paper 9, one could still describe the next step as a generic expansion toward phenomenality. After Paper 9, that description is no longer adequate. The next step is now

a typed scientific program with concrete ways to fail, concrete reasons for those failures, and concrete remediation paths. That is a substantive change in the research program even before bridge admissibility has been achieved, because it transforms an open-ended aspiration into a burdened formal agenda.

# S Interpretation and Open Research Links

**Historical alias.** Phenomenal Bridge Organization. The alias is retained for bibliographic continuity; the technical title states the demonstrated object.

**Formal anchor.** Results 6.13 and 9.1.

**Proposed interpretation.** The phenomenal-bridge alias proposes a further connection to experience. The proved interaction and transport results describe mathematical comparison problems.

**Additional obligations.** Local certificates must still be linked to the full predicate stack, broad rival closure and independently identified observables.

**Differential tests and evidence.** Retain the exact interacting-rival tie and information-access controls. Test new predictions out of sample under the same complexity convention.

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